cog-SUP Master 2 Thesis

Track: Computational Linguistics and AI

Toward a Shared Semantic Hub: Evidence from Aligning Multilingual LLMs with *Le Petit Prince* fMRI

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Declaration of originality

This thesis is an original research project that extends previous work on language-model-based fMRI encoding to a multilingual setting. While previous analyses of the *Le Petit Prince* fMRI dataset mainly focused on English, the present study compares model–brain alignment across English, French, and Chinese, and asks whether encoding models trained in one language can generalize to brain responses in another language.

The originality of the project lies in three main aspects. First, it constructs ISC-matched average subjects across the three languages, in order to reduce differences in fMRI reliability before comparing model–brain alignment. Second, it evaluates both within-language prediction and strict cross-language transfer using several large language models, including an English-dominant model and recent multilingual models. Third, it complements transfer analyses with beta-pattern similarity analyses, allowing the comparison of model-to-brain mappings across languages.

By combining reliability-controlled fMRI targets, multilingual encoding models, cross-language transfer, and encoding-weight similarity, this thesis aims to go beyond existing work by testing whether multilingual LLM representations support partially shared brain alignment across languages. The results are not intended to prove the existence of a fully language-independent semantic system, but they can provide evidence relevant to the open question of how shared and language-specific representations interact in multilingual model–brain alignment.

Declaration of contribution

Defining the scientific question:
Laurent Bonnasse-Gahot and Christophe Pallier, with contributions from Xinyu Zhou.

Bibliographic research to define the question and guide the choice of approach and methodology:
Xinyu Zhou, with guidance from Laurent Bonnasse-Gahot and Christophe Pallier.

Choosing the general approach to answer the question:
Laurent Bonnasse-Gahot, Christophe Pallier, and Xinyu Zhou.

Choosing and developing the specific methodology:
Xinyu Zhou, with guidance and feedback from Laurent Bonnasse-Gahot and Christophe Pallier.

Developing the analysis plan:
Xinyu Zhou, Laurent Bonnasse-Gahot, and Christophe Pallier.

Data collection:
The fMRI data used in this thesis come from an existing trilingual *Le Petit Prince* dataset and were not collected by Xinyu Zhou during this internship.

Data organization and preprocessing:
Xinyu Zhou, with methodological guidance from Laurent Bonnasse-Gahot and Christophe Pallier.

Programming the analyses:
Xinyu Zhou.

Coding and data analysis:
Xinyu Zhou.

Interpreting the results and formulating conclusions:
Xinyu Zhou, with feedback from Laurent Bonnasse-Gahot and Christophe Pallier.

Writing the dissertation:
Xinyu Zhou.

Proofreading and detailed feedback on the dissertation:
Laurent Bonnasse-Gahot and Christophe Pallier.

Abstract

Understanding whether different languages rely on shared neural representations is a central question in cognitive neuroscience and multilingual language modelling. This thesis investigates this question by aligning large language model representations with fMRI responses collected during naturalistic story comprehension in English, French, and Chinese. Using the trilingual *Le Petit Prince* fMRI dataset, we examine whether model–brain mappings learned in one language can generalize to another language, and whether multilingual large language models provide more stable cross-language alignment than an English-dominant model.

A major methodological challenge is that the three language groups contain imbalanced numbers of participants and may therefore differ in fMRI reliability, which would confound direct cross-language comparisons. To reduce this confound, we first constructed inter-subject correlation (ISC)-matched average subjects. English and Chinese participant groups were selected from larger pools to match the fixed French group in both participant number and inter-subject correlation. The final groups showed highly similar scalar ISC values and similar voxel-wise ISC distributions, providing more comparable neural targets for subsequent encoding analyses.

We then fitted ridge-regression encoding models using layer-wise activations from an English-dominant model (GPT-2 XL) and two multilingual models (Llama 3.1 8B and Qwen3 8B). Model activations were aligned with audiobook timing, convolved with a canonical haemodynamic response function, and used to predict average-subject fMRI responses. Within-language results showed a clear English advantage for the English-dominant model, whereas the two multilingual models achieved more balanced prediction scores across English, French, and Chinese. Cross-language transfer analyses further showed that the English-dominant model suffered a large transfer loss, especially for conditions involving Chinese. In contrast, the two multilingual models showed much smaller differences between within-language prediction and cross-language transfer. Finally, beta-pattern similarity analyses showed that the multilingual models learned more similar model-to-brain mappings across languages than the English-dominant model, particularly in middle layers.

Overall, these results suggest that cross-language model–brain alignment is feasible and is stronger for multilingual models than for an English-dominant model. The findings are consistent with the existence of partially shared cross-language representations during naturalistic story comprehension, while leaving open the relative contributions of semantic, syntactic, and lower-level linguistic information.

Keywords: model–brain alignment; multilingual large language models; fMRI; inter-subject correlation; cross-language transfer; semantic representations; *Le Petit Prince*

Pre-registration document

vi **Pre-registration: Cross-language Transfer and Shared Representations in Multilingual LLM–Brain Alignment**

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Scientific Background

Human language comprehension relies on a left-lateralized fronto-temporal network that is robust across typologically diverse languages [1]. A functional differentiation within this network has been repeatedly suggested: inferior frontal and posterior temporal regions are relatively more sensitive to syntactic/formal structure, whereas more anterior/inferior frontal and temporo-parietal regions (e.g. BA45/pSTS/AG) contribute more strongly to semantic and combinatorial meaning [2, 3].

Multilingual language models learn partially shared representational spaces across languages [4, 5]. Recent work further suggests a layered organization where intermediate representations become more concept-/meaning-oriented and less tied to surface form [6]. In parallel, model–brain alignment studies show that representations from neural language models can be linearly mapped onto fMRI responses during naturalistic comprehension [7–9].

A central focus of this project is **cross-language transfer in model–brain alignment**: to what extent can an encoding model trained in one language generalize zero-shot to another, and what does successful transfer imply about **shared representational geometry** across languages? Recent multilingual fMRI work provides evidence that encoding models can transfer across many languages, but systematic characterization of *when* transfer succeeds/fails and *what* model components support transfer remains limited [10]. Mechanistic studies of multilingual LLMs identify language-specific versus language-invariant components at the neuron/subspace level (e.g. via LAPE), offering concrete tools to probe which components are most predictive of cross-language transfer [11].

Research Questions

Primary questions (transfer and shared representations):

- How well do LLM→fMRI encoding models transfer across languages (EN/FR/ZH) under naturalistic comprehension?
- Is transfer asymmetric across language pairs and model families (multilingual vs monolingual baselines; layer dependence)?
- How do participant selection, averaging strategy, and masking/ROI choices constrain ISC ceilings and transfer estimates?

Secondary questions (characterizing what transfers):

- Do language-specific vs language-invariant LLM units/subspaces differentially contribute to cross-language transfer?
- (Exploratory) Can form–meaning sensitivity (syntax/semantics) help interpret which components enable transfer and where they align in the language network?

Experimental Conditions

Neuroimaging condition (core). Brain responses are measured during naturalistic auditory comprehension of *Le Petit Prince* in the participant's native language (English/French/Chinese). This enables evaluating shared representations and transfer under ecologically valid comprehension.

Optional linguistic manipulation (exploratory). If time permits, I will use linguistically controlled minimal pairs from MultiBLiMP [12] to derive diagnostic contrasts (syntax-dominant vs semantics-dominant). These contrasts can be used to probe whether transfer-relevant components are more form- or meaning-associated.

Datasets

- **fMRI:** Multilingual *Le Petit Prince* dataset[13].
- **Optional minimal pairs:** MultiBLiMP subsets in EN/FR/ZH (syntax- vs semantics-dominant phenomena) [12].

Models

Decoder-only Transformers:

- Multilingual: Qwen family, Mistral family, Llama family
- Bilingual: CroissantLLM
- Monolingual baseline: GPT-2 family

Layer-wise hidden states will be extracted, aligned to the audiobook transcript timing, and HRF-convolved.

Measurements and Analyses

ISC (data quality / ceiling). Whole-brain and ROI-wise ISC will be computed with equalized participant counts across languages. Comparisons include random vs quiz-based participant selection (top vs bottom comprehension where available), to quantify ceilings and reliability constraints for transfer.

Encoding models (LLM→fMRI): within-language and cross-language transfer (core). Ridge regression maps HRF-convolved LLM features to voxel-wise BOLD. Brain scores are Pearson correlations between predicted and observed signals. Analyses include within-language fits and cross-language zero-shot transfer (reusing weights) across EN/FR/ZH, with layer-wise and ROI-wise evaluations.

What transfers? (mechanistic / feature-level). Language Activation Probability Entropy (LAPE) will identify language-specific vs language-invariant neurons. Feature subsets (low- vs high-LAPE) will be tested for differential contributions to within-language fits vs cross-language transfer, aiming to characterize components supporting shared representations.

Optional exploratory: syntax–semantics lens. If feasible, minimal-pair-derived contrasts will be used to construct syntax-sensitive vs semantics-sensitive predictors/directions, to examine whether transfer-relevant components preferentially align with form-sensitive vs meaning-sensitive ROIs.

Timeline

- Weeks 1–3: ISC analyses; participant equalization; mask/ROI strategy comparison; cross-language pipeline sanity checks
- Weeks 4–7: Core encoding models: within-language fits + cross-language transfer (weights reuse); layer/model comparisons
- Weeks 8–10: Transfer diagnostics: ROI-wise summaries; robustness to averaging, masking, and participant selection
- Weeks 11–13: Mechanistic analysis: LAPE-based language-specific vs invariant components; feature-subset transfer tests
- Weeks 14–16: Optional minimal-pair probing (MultiBLiMP) as an interpretive lens (syntax/semantics)
- Weeks 17–20: Integration, visualization, and thesis writing

Expected Contributions

This project will (i) provide a systematic evaluation of **cross-language transfer** in multilingual LLM→fMRI encoding models across EN/FR/ZH under naturalistic comprehension, (ii) clarify how **data reliability/ceilings** (ISC) and analysis choices constrain transfer estimates, and (iii) characterize **which model components support transfer** (language-invariant vs language-specific; LAPE), with optional exploratory interpretation via syntax/semantics.

References

[1] Saima Malik-Moraleda, Dima Ayyash, Jeanne Gallée, Josef Affourtit, Malte Hoffmann, Zachary Mineroff, Olessia Jouravlev, and Evelina Fedorenko. An investigation across 45 languages and 12 language families reveals a universal language network. *Nature Neuroscience*, 25(8):1014–1019, 2022. doi: 10.1038/s41593-022-01114-5.

[2] Christophe Pallier, Anne-Dominique Devauchelle, and Stanislas Dehaene. Cortical representation of the constituent structure of sentences. *Proceedings of the National Academy of Sciences*, 108(6):2522–2527, 2011. doi: 10.1073/pnas.1018711108.

[3] Tomás Goucha and Angela D. Friederici. The language skeleton after dissecting meaning: A functional segregation within broca's area. *NeuroImage*, 114:294–302, 2015. doi: 10.1016/j.neuroimage.2015.04.011.

[4] Telmo Pires, Eva Schlinger, and Dan Garrette. How multilingual is multilingual BERT? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4996–5001, Florence, Italy, 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1493. URL `https://aclanthology.org/P19-1493/`.

[5] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.747. URL `https://aclanthology.org/2020.acl-main.747/`.

[6] Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. Do llamas work in english? on the latent language of multilingual transformers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15366–15394, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.820. URL `https://aclanthology.org/2024.acl-long.820/`.

[7] Charlotte Caucheteux and Jean-Rémi King. Brains and algorithms partially converge in natural language processing. *Communications Biology*, 5:134, 2022. doi: 10.1038/s42003-022-03036-1.

[8] Martin Schrimpf, Idan A. Blank, Greta Tuckute, Carina Kauf, Elika A. Hosseini, Nancy Kanwisher, Joshua B. Tenenbaum, and Evelina Fedorenko. The neural architecture of language: Integrative modeling converges on predictive processing. *Proceedings of the National Academy of Sciences*, 118(45):e2105646118, 2021. doi: 10.1073/pnas.2105646118.

[9] Laurent Bonnasse-Gahot and Christophe Pallier. fmri predictors based on language models of increasing complexity recover brain left lateralization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL `https://arxiv.org/abs/2405.17992`.

[10] Andrea Gregor de Varda, Saima Malik-Moraleda, Greta Tuckute, and Evelina Fedorenko. Multilingual computational models capture a shared meaning component in brain responses across 21 languages. *bioRxiv*, 2025. doi: 10.1101/2025.02.01.636044. Preprint (version 2 posted 2025-11-18).

[11] Tianyi Tang, Wenyang Luo, Haoyang Huang, Dongdong Zhang, Xiaolei Wang, Xin Zhao, Furu Wei, and Ji-Rong Wen. Language-specific neurons: The key to multilingual capabilities in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5701–5715, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.309. URL `https://aclanthology.org/2024.acl-long.309/`.

[12] Jaap Jumelet, Leonie Weissweiler, and Arianna Bisazza. Multiblimp 1.0: A massively multilingual benchmark of linguistic minimal pairs. *arXiv*, 2025. doi: 10.48550/arXiv.2504.02768. URL `https://arxiv.org/abs/2504.02768`.

[13] Jixing Li, Shohini Bhattasali, Shulin Zhang, Berta Franzluebbers, Wen-Ming Luh, R Nathan Spreng, Jonathan R Brennan, Yiming Yang, Christophe Pallier, and John Hale. Le petit prince multilingual naturalistic fmri corpus. *Scientific data*, 9(1):530, 2022.

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Introduction

Human languages differ substantially in phonology, morphology, syntax, and writing systems. French marks grammatical gender, Chinese uses lexical tones, and English relies on a different combination of word order, morphology, and prosody. Despite this diversity, neuroimaging studies have consistently identified a broadly similar left-lateralized fronto-temporo-parietal language network involved in language comprehension across many languages Hertrich, Dietrich, and Ackermann (2020); Lin, Zhang, Wang, and Wang (2024); Malik-Moraleda et al. (2022a); C. J. Price (2012); Vigneau et al. (2006). This suggests that the human brain may rely on a partially shared neural architecture for language comprehension.

However, similar anatomical localization does not necessarily imply identical neural representations. Different languages may activate overlapping cortical regions while still being encoded through language-specific representational patterns. This distinction is especially important when studying meaning. Semantic comprehension appears to rely on high-level cortical systems that are relatively stable across languages, whereas phonological, lexical, syntactic, and morphological processing may introduce language-specific modulations Malik-Moraleda et al. (2022a); Nieuwland, Martin, and Carreiras (2012); Wei et al. (2023). The central question is therefore not only whether different languages recruit similar brain areas, but whether they give rise to shared representational structure in the brain. In other words, is there a cross-linguistically stable neural code for meaning during natural language comprehension?

Multilingual large language models (LLMs) provide a useful computational framework for addressing this question. These models are trained on text from multiple languages and often develop high-dimensional representations in which semantically equivalent sentences are closer to one another, even when they are expressed in different languages Qin et al. (2025). Recent work on multilingual model interpretability suggests that this organization is not uniform across layers. Early and late layers tend to contain more language-specific information, such as script, lexical form, and output-language control, whereas middle layers often show stronger cross-lingual semantic alignment Bhattacharya and Bojar (2023); Lindsey et al. (2025); Z. Wu, Yu, Yogatama, Lu, and Kim (2024). This has led to the hypothesis that multilingual models may contain a partially shared semantic or conceptual space, while still preserving language-specific interfaces.

Model–brain alignment offers a way to connect these computational representations with neural data. In this framework, representations extracted from language models are used to predict brain responses recorded during language comprehension. Previous studies have shown that contextual representations from modern language models can predict fMRI, MEG, or ECoG responses to words, sentences, and narratives Caucheteux and King (2022). In multilingual settings, this approach can be extended to ask whether a model-to-brain mapping learned in one language can generalize to another language. Successful cross-language transfer would suggest that at least part of the model–brain correspondence reflects shared representational structure rather than only language-specific surface form.

Recent multilingual encoding studies provide evidence in this direction. de Varda, Malik-Moraleda, Tuckute, and Fedorenko (2025) examined model–brain alignment across 21 languages and showed that multilingual language-model representations can predict fMRI responses across typologically diverse languages. They further found that alignment was related to next-word-prediction performance, model capacity, and training objective. In a closely related study, Zada, Nastase, Li, and Hasson (2025) used naturalistic fMRI responses to the same story in English, French, and Chinese, and showed that language-model embeddings trained or evaluated in one language can generalize to brain responses elicited by another language. Their results suggest that cross-language model–brain alignment is strongest in high-level language and default-mode regions, consistent with a shared conceptual representation across languages.

However, these studies also leave open several questions. Some analyses relied on relatively small numbers of participants per language, while others focused mainly on BERT-style or earlier multilingual models. It therefore remains unclear how recent larger decoder-only models behave in a controlled cross-language encoding setting, and whether their model-to-brain mappings are more stable across languages than those of an English-dominant model.

Present study. The present thesis investigates cross-language model–brain alignment using the trilingual *Le Petit Prince* fMRI dataset, in which native speakers of English, French, and Chinese listened to the same narrative in their own language. This dataset allows us to test whether large language model representations can predict brain responses across languages that differ substantially in linguistic form and writing system.

To make the comparison more reliable, this thesis first constructs ISC-matched average subjects. English and Chinese participant groups are selected to match the fixed French group in both participant number and whole-brain inter-subject correlation, reducing potential confounds due to differences in fMRI reliability.

Using these matched neural targets, the thesis compares GPT-2 XL with two recent multilingual models, Llama 3.1 8B and Qwen3 8B. The analyses examine within-language prediction, cross-language transfer, and similarity between language-specific encoding weights. The goal is to test whether multilingual models provide stronger evidence for partially shared cross-language model–brain mappings than an English-dominant model.

Chapter 1

Related Work

1.1 Model–brain alignment of Multilinguality

1.1.1 From model-brain alignment to language comprehension

Model-brain alignment studies examine the extent to which representations derived from computational models can predict, explain, or resemble neural responses elicited by corresponding cognitive or perceptual processes Kriegeskorte, Mur, and Bandettini (2008); Naselaris, Kay, Nishimoto, and Gallant (2011). In the language domain, model-brain alignment studies examine the extent to which representations derived from computational models of language can predict, explain, or resemble neural responses elicited during language-related tasks Mitchell et al. (2008). In monolingual settings, a large body of work has shown that contextualized representations from modern language models can be linearly mapped onto fMRI, MEG, or ECoG responses during sentence or narrative processing. For instance, Caucheteux and King (2022) showed that deep language algorithms partially converge toward brain-like representations, and that this convergence is closely related to their ability to predict words from context. Oota, Gupta, and Toneva (2023) further intervene on surface, syntactic, and semantic properties in model representations, showing that different linguistic properties contribute differently to model-brain alignment across layers. These studies establish model-brain alignment as a useful framework for asking not only whether models resemble the brain, but also which levels of linguistic representation support this resemblance.

1.1.2 Evidence for shared semantic representations across languages

In multilingual contexts, the central question becomes whether this alignment is language-specific or whether it reflects a shared representational space for meaning across languages. Earlier bilingual neuroscience studies already suggest that the human brain can encode meaning beyond the surface form of a particular language. Honey, Thompson, Lerner, and Hasson (2012) showed that listeners processing the same narrative in different languages exhibit shared neural responses in high-level cortical areas, suggesting that narrative-level comprehension can generalize across languages. Correia et al. (2014) provided converging evidence from bilingual fMRI decoding, showing that spoken words in different languages can evoke language-independent semantic representations in the anterior temporal lobe. Xu, Li, and Li (2021) further framed cross-language brain decoding as a direct way to test whether neural representations trained in one language can generalize to another. Malik-Moraleda et al. (2022b) investigated the fronto-temporo-parietal language network across 45 languages and showed that its broad topography and key functional properties are robust to cross-linguistic variation. These findings motivate the hypothesis that the brain may contain a partially language-general semantic system, while still preserving language-specific processing for phonological, lexical, and syntactic form.

1.1.3 Cross-lingual transfer in multilingual brain encoding

Recent studies have connected this bilingual neuroscience tradition with language models. In bilingual readers or listeners, semantic representations across L1 and L2 appear to be largely shared but not identical. Hou, Li, Gao, Ou, and Xu (2024) showed that syntactic and semantic information in Chinese-English bilinguals recruits overlapping but distinguishable neural patterns across languages. Gu, Nastase, Zada, and Li (2025) used LLM-derived contextual embeddings to model L1 and L2 reading comprehension and found broadly comparable model-brain alignment across languages, while also showing that L2 alignment is more strongly modulated by individual differences such as attention and language dominance. Similarly, Chen et al. (2026) reported that bilingual semantic representations are shared across languages but systematically modulated by each language. Taken together, these findings support a "shared core with language-specific modulation" account rather than a strictly language-independent account.

More direct evidence for multilingual model-brain alignment comes from recent work using many languages and multilingual computational models. de Varda et al. (2025) showed that multilingual models such as XLM-R and mBERT can capture a shared meaning component in brain responses across 21 languages, and that encoding models can generalize zero-shot across languages. This suggests that at least part of the correspondence between models and brains reflects language-general meaning rather than language-specific form. Zada et al. (2025) provided closely related evidence using naturalistic fMRI responses to the same story in English, Chinese, and French. They found that language models trained on different languages converge toward a similar embedding space, especially in middle layers, and that voxel-wise encoding models trained on one language can predict neural responses in listeners processing the same content in another language, particularly in high-level language and default-mode regions.

These studies jointly suggest that multilingual model-brain alignment is organized around a partially universal semantic space. However, the evidence does not imply that all language processing is language-agnostic. Cross-lingual generalization is strongest in higher-level semantic and narrative regions, while lower-level perceptual, lexical, and language-specific structures may remain more dependent on the input language. This distinction is crucial for the present study: if multilingual LLMs and human brains both contain a shared semantic hub, then model-to-brain alignment should transfer across languages above chance; if language-specific interfaces remain important, then transfer performance should vary depending on the source language, target language, model architecture, and brain region.

1.2 Interpretability of Multilingual LLMs

1.2.1 Cross-lingual transfer and shared multilingual representations

Interpretability research on multilingual LLMs provides a complementary perspective on why cross-lingual model-brain alignment may be possible. Earlier studies on multilingual encoders such as mBERT showed that multilingual pretraining can give rise to shared representations that support zero-shot transfer. Pires, Schlinger, and Garrette (2019) found that mBERT transfers surprisingly well across languages, even across different scripts, though transfer remains stronger between typologically similar languages. S. Wu and Dredze (2019) similarly showed that multilingual BERT is effective across many languages, suggesting that it learns representations that are not simply language-specific. Conneau, Wu, Li, Zettlemoyer, and Stoyanov (2020) further demonstrated that cross-lingual structure can emerge even without a shared vocabulary, as long as higher layers share parameters across languages. At the syntactic level, Chi, Hewitt, and Manning (2020) found that mBERT encodes universal grammatical relations in partially shared syntactic subspaces. Libovický, Rosa, and Fraser (2020) also showed that multilingual contextual embeddings are partially language-neutral, especially for lexical-semantic information.

1.2.2 Language-specific neurons and features

With the rise of decoder-style LLMs, recent work has shifted from broad cross-lingual transfer to more mechanistic questions: where are languages represented inside the model, and how are shared and language-specific computations organized? A consistent finding is that multilingual LLMs contain language-selective components, especially in early and late layers. Bhattacharya and Bojar (2023) showed that feed-forward networks in multilingual Transformers contain language-specific specialization. Kojima, Okimura, Iwasawa, Yanaka, and Matsuo (2024) identified neurons that fire selectively for individual languages in decoder-based multilingual models; these neurons are mainly located in the first and last layers, and manipulating fewer than 1% of them can strongly affect the probability of generating a target language. Tang et al. (2024) proposed Language Activation Probability Entropy (LAPE) to identify language-specific neurons and showed that selectively activating or deactivating these neurons can steer the output language. Zhang, Zhao, Zhang, Gui, and Huang (2024) extended this view by identifying both core linguistic regions and distinct monolingual regions inside LLMs.

These findings indicate that multilingual LLMs are not fully language-agnostic systems. Instead, they preserve language-specific interfaces that handle input and output forms. However, language-specific components coexist with shared computation. Choenni, Garrette, and Shutova (2023); Choenni, Shutova, and Garrette (2024) showed that language-specialized subnetworks can arise in multilingual models and influence the balance between cross-lingual sharing and interference. This supports a hybrid architecture in which some components are specialized for particular languages, while other components are shared across languages and tasks. This organization matches the structure highlighted in recent multilingual interpretability surveys: multilingual LLMs contain both language-selective components and shared representations that support cross-lingual transfer and semantic alignment.

1.2.3 Layer-wise organization: from language-specific interfaces to shared concept spaces

Another line of work probes intermediate representations directly and provides stronger evidence for a shared concept space. Wendler, Veselovsky, Monea, and West (2024), using the logit lens on Llama-2 models, proposed that multilingual Transformers pass through three representational phases: an input-language space, a middle-layer concept space, and an output-language space. Importantly, they found that the middle-layer concept space in English-dominant models is often closer to English than to the input language, suggesting an English-biased latent pivot rather than a perfectly language-neutral semantic space. Dumas, Wendler, Veselovsky, Monea, and West (2025) strengthened this claim through activation patching, showing that concept information and output-language information can be partly separated: one can change the concept while preserving the output language, or change the output language while preserving the concept. They further showed that averaging latent representations across languages can preserve or even improve translation performance, supporting the existence of language-agnostic concept representations.

Other recent studies converge on a similar view. Zeng, Han, Chen, and Yu (2025) proposed that multilingual LLMs develop a hidden-state "lingua franca," a shared semantic latent space into which semantically equivalent inputs from different languages are mapped; this alignment becomes stronger with training and model scale. Liu et al. (2024) found shared multilingual activation patterns and language-family-related structure in LLMs. Ferrando and Costa-Jussà (2024) showed that circuits for subject-verb agreement can be similar across languages, suggesting that some grammatical computations are reused cross-linguistically. Sparse-autoencoder-based studies further refine this picture: Deng, Wan, Yang, Zhang, and Feng (2025) found that some SAE features are strongly language-specific and that ablating them selectively harms one language while leaving others relatively unaffected. These results suggest that language-specificity may be better understood not only at the neuron level, but also at the level of sparse, interpretable features.

Overall, multilingual LLM interpretability supports a hybrid account: early and late layers tend to encode language-specific forms and output control, whereas middle layers are more likely to encode abstract semantic or conceptual content shared across languages. This account provides a mechanistic explanation for multilingual model-brain alignment. If middle layers contain shared semantic representations, they should be better candidates for cross-lingual brain alignment, especially in high-level semantic regions. Conversely, if language-specific neurons and features dominate lower and upper layers, alignment in these layers may be more sensitive to the source and target languages. The present study builds on this connection by asking whether cross-lingual transfer in model-brain encoding reflects the same shared semantic hub suggested by multilingual LLM interpretability, and whether deviations from transfer reveal language-specific organization in either models or brains.

Chapter 2

Inter-subject Correlation and ISC-matched Average Subjects

2.1 Background and motivation

This thesis builds on the framework of Bonnasse-Gahot and Pallier (2024). Their main analyses focused on the English part of the *Le Petit Prince* fMRI dataset. Using fMRI data already spatially normalized to MNI space, they resampled the data, constructed a common brain mask, averaged the responses across participants to obtain an average subject, and then fitted encoding models to this group-level fMRI target. They also estimated inter-subject reliability independently of any language model, using a more elaborate split-half inter-group prediction procedure: subjects were divided into two subgroups, the average response of one subgroup was used to predict the average response of the other subgroup through ridge regression, and voxel-wise prediction correlations were computed across held-out runs.

In the present thesis, we use a simpler ISC estimate: for each language group, subjects are split into two equal halves, the average BOLD response is computed within each half, and voxel-wise Pearson correlations are computed directly between the two half-group average time series. This differs from the ridge-regression-based inter-group prediction procedure used by Bonnasse-Gahot and Pallier (2024), but serves the same purpose of estimating language-model-independent reliability of the fMRI target.

The present thesis also extends this idea to a multilingual setting. We compare model–brain alignment across English, French, and Chinese, and we ask whether a mapping learned in one language can transfer to another language. This comparison requires not only the same encoding model, but also comparable fMRI targets. If one language has a cleaner group-level brain response than another language, then a higher brain score may simply reflect higher signal reliability, not a better linguistic match between model and brain. This issue is especially important in the multilingual *Le Petit Prince* dataset. The available participant pools are unbalanced: 49 English participants, 35 Chinese participants, and 28 French participants. If the average subject were built from all available participants in each language, the English target would be based on more participants than the French target. The English average subject would therefore be expected to have a higher signal-to-noise ratio. Directly comparing brain scores across languages would then be difficult.

To reduce this confound, we used inter-subject correlation (ISC) to estimate the reliability of the fMRI response in each language before fitting encoding models. ISC measures how similarly different participants' brains respond to the same time-locked stimulus. In naturalistic story listening, participants hear the same narrative over time. If a voxel is reliable, its time series should be similar across independent groups of listeners. ISC is useful here because it does not require a language model. It measures the reproducibility of the brain data itself.

This chapter describes two steps. First, we report exploratory ISC analyses showing that group size and listener proficiency affect the reliability of the average brain. Second, we describe the final ISC-matching procedure used to construct matched average subjects for English, French, and Chinese. These average subjects are matched in participant number and in whole-brain ISC. They are then used as the fMRI targets in the encoding and cross-language transfer analyses.

2.2 Dataset and preprocessing

In multilingual *Le Petit Prince* fMRI dataset, participants listened to the audiobook of *Le Petit Prince* in their native language. The dataset contains English, French, and Chinese groups. Each participant was scanned during 9 runs, with a repetition time of 2 s. The run structure is comparable across languages, which makes the dataset suitable for cross-language analyses of naturalistic language comprehension.

The participant pools are unequal in size: 49 English participants, 35 Chinese participants, and 28 French participants. Since French is the smallest group, we used 28 participants as the target group size. In the final matched dataset, all 28 French participants were kept, while 28 English and 28 Chinese participants were selected from the larger pools.

All analyses used a common brain mask across all participants. In the final implementation, this mask contained 26,439 voxels. Using a common mask ensures that ISC scores and average subjects are computed in the same voxel space for English, French, and Chinese.

For ISC matching, each subject's fMRI data were transformed with `NiftiMasker`. The masker applied the common mask, linear detrending, standardization, high-pass filtering with a cut-off of $1/128$ Hz, and 8 mm FWHM smoothing. The first and last 10 TRs of each run were removed before ISC estimation, because run boundaries may contain transition artifacts. After preprocessing, each language was represented by 9 run-level matrices with dimensions:

$$N_{subjects} \times T_{run} \times V, \tag{2.1}$$

where $N_{subjects}$ is the number of participants, T_{run} is the number of time points in the run after trimming, and V is the number of voxels in the common mask.

The preprocessing used for ISC matching and the preprocessing used for the final average subjects were closely related but not identical. Smoothing and trimming were used during ISC matching to obtain stable reliability estimates for group selection. After the matched groups were selected, the final average subjects were rebuilt with the preprocessing expected by the downstream encoding pipeline.

2.3 Group-wise inter-subject correlation

We used a split-half group-wise ISC. For a given language and a given group of participants, the group is randomly divided into two equal halves. The fMRI signals are averaged within each half. Then, for each voxel, we compute the Pearson correlation between the two half-group average time series. This gives one voxel-wise ISC value. The whole-brain ISC score is obtained by averaging these voxel-wise values across voxels and across the 9 runs.
For one split s of a group G, the scalar ISC score can be written simply as:

$$ISC_s(G) = \frac{1}{KV} \sum_{k=1}^{K} \sum_{v=1}^{V} r_{s,k,v}, \tag{2.2}$$

where $K = 9$ is the number of runs, V is the number of voxels, and $r_{s,k,v}$ is the correlation between the two half-group average time series for run k and voxel v.
A high ISC means that two independent halves of the group show similar time courses. A low ISC means that the group-level response is noisier or less consistent across participants.

2.4 Exploratory ISC analyses

Before constructing the final matched average subjects, we first examined two factors that could affect group-wise ISC. The first factor was sample size: larger groups should produce more reliable average responses because individual noise is reduced by averaging. The second factor was comprehension quiz score: participants who understood the story better may show more synchronized neural responses during naturalistic listening.

2.4.1 Effect of sample size

The first analysis tested whether ISC increases when more participants are averaged. For each language, we sampled random groups from the full participant pool. The half-group sizes were $n = 10$, $n = 12$, and $n = 14$, corresponding to total group sizes of 20, 24, and 28 participants. For each language and each group size, we ran 30 independent resampling iterations with controlled random seeds. For each resample, participants were split into two halves, the two half-group responses were averaged, and voxel-wise ISC was computed across the 9 runs.
Whole-brain ISC increased with group size in all three languages. In Chinese, ISC increased from 0.246 at $n = 10$ to 0.304 at $n = 14$. In English, ISC increased from 0.223 at $n = 10$ to approximately 0.26 at $n = 14$. French showed the same trend and reached approximately 0.29 at the largest group size. This confirms that averaging more participants reduces individual noise and produces a more stable group-level response.
This result directly motivates the matched-average-subject procedure. If participant number affects ISC, then comparing average subjects built from different numbers of participants would introduce a confound. A language with more participants would have a more reliable target, and this could inflate its model–brain alignment score.

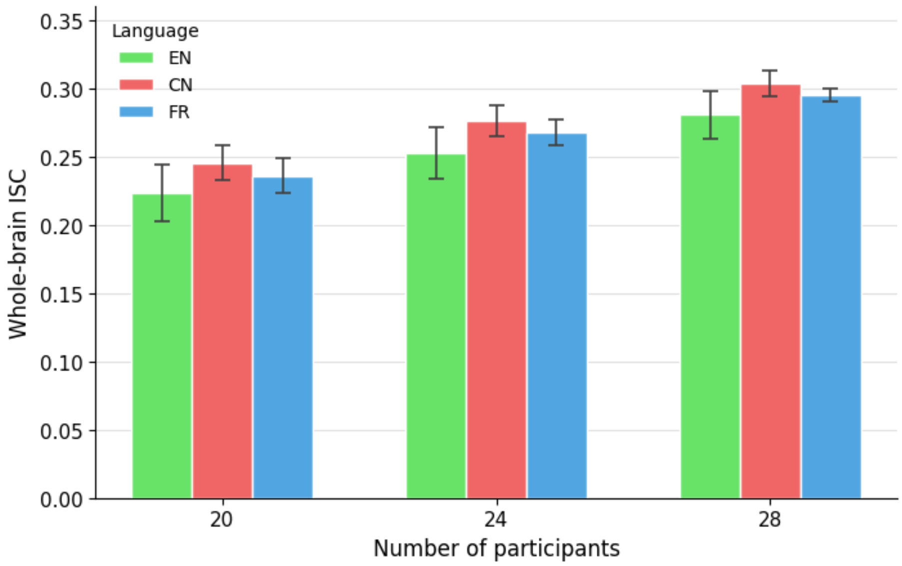

Figure 2.1: **Effect of group size on whole-brain ISC.** Random participant groups were sampled in English, French, and Chinese. The x-axis shows the number of participants in each split half ($n = 10, 12, 14$), corresponding to total group sizes of 20, 24, and 28. Error bars show standard deviation across resampling iterations.

2.4.2 Effect of comprehension quiz score

The second analysis tested whether behavioral comprehension scores were related to neural synchrony. In the released participant metadata, comprehension quiz scores are available for the English and Chinese participants, but not for the French participants. We therefore performed this analysis only for English and Chinese.

Participants in the English and Chinese experiments answered four-choice comprehension questions after each section of the audiobook, with 36 questions in total. The number of correct answers was used as a simple behavioral measure of story comprehension. For each language, participants were ranked by quiz score. We then compared three types of groups: participants with the highest quiz scores, participants with the lowest quiz scores, and randomly selected participants.

For each half-group size n, the top $2n$ participants and bottom $2n$ participants were selected according to their quiz scores. Within each selected group, participants were randomly split into two halves of size n. Group-wise ISC was then computed between the two half-group average responses. This was repeated across resampling iterations. A random group of the same total size was used as a reference.

In English, groups with higher quiz scores showed higher ISC for small and medium group sizes. At $n = 12$ (24 participants in total), the high-score group reached an ISC of 0.309 ± 0.004, whereas the low-score group reached 0.208 ± 0.005, with the random group falling in between. As group size increased, this difference became smaller and was no longer visible when almost all English participants were included.

The Chinese data showed the same qualitative pattern. High-score groups produced higher ISC than low-score groups, especially for small and medium group sizes. At $n = 12$, the high-score group reached an ISC of 0.296 ± 0.005, compared with 0.259 ± 0.003 for the low-score group. At $n = 16$, the gap became smaller. Because the Chinese pool contained

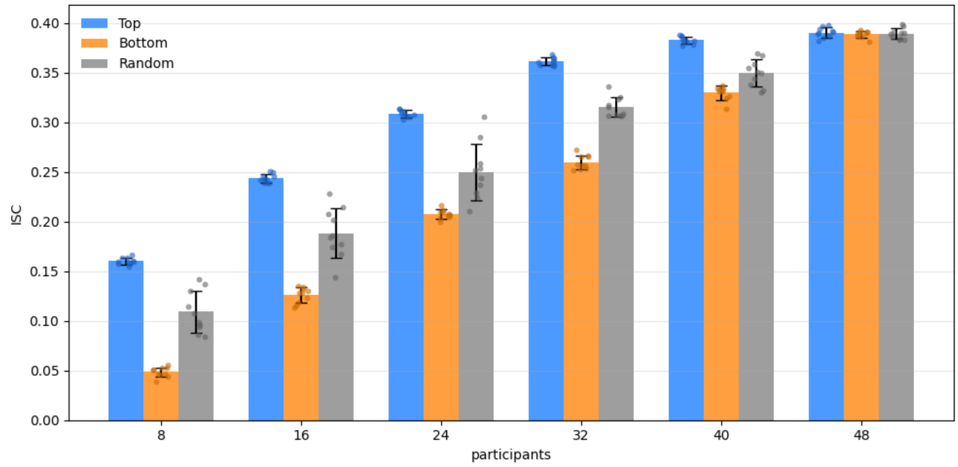

Figure 2.2: **Effect of comprehension quiz score on English ISC.** English participants were ranked by the number of correctly answered comprehension questions. For each group size, ISC was computed for the top-score group, the bottom-score group, and a random reference group.

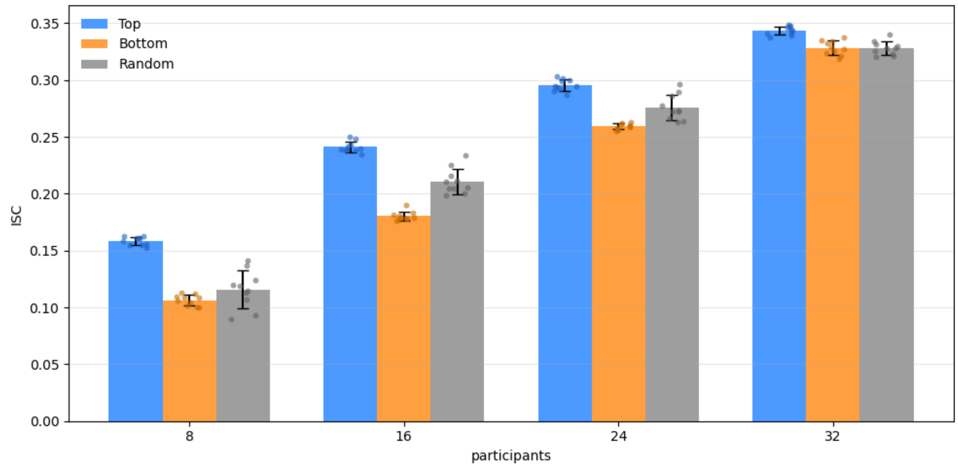

Figure 2.3: **Effect of comprehension quiz score on Chinese ISC.** Chinese participants were ranked by the number of correctly answered comprehension questions.

35 participants, the analysis could not be extended to the larger English group sizes of 40 and 48 participants.

The full numerical summaries for both languages are reported in Appendix A.1 and Appendix A.2.

These results suggest that ISC is related not only to the number of participants being averaged, but also to behavioral comprehension during the task. Participants who answered more comprehension questions correctly tended to show more synchronized neural responses, especially when the group size was small.

2.5 ISC-matched average subjects

The downstream encoding analyses fit language-model predictors to fMRI time series. For each language and each run, the fMRI time series are averaged across a selected group of participants. This produces one group-level fMRI target per language. The average subjects must be constructed in a comparable way. If one language average is based on more participants, or on a group with higher inter-subject reliability, it may be easier to predict. For this reason, we did not build the English, Chinese, and French average subjects from all available participants. Instead, we constructed ISC-matched average subjects. The goal was to use the same number of participants in each language and to match the inter–subject correlation of the group-level fMRI response before running the encoding analyses.

2.5.1 Group selection

French had the smallest participant pool, with 28 participants, and was therefore used as the reference group. All 28 French participants were kept. English and Chinese had larger participant pools, with 49 and 35 participants respectively. For these two languages, 28 participants were selected from the larger pools.

The selection was based on repeated split-half ISC. For English and Chinese, we randomly sampled 50 candidate groups of 28 participants. Each candidate group was evaluated with 10 repeated random split-half estimates. In each split, the 28 participants were divided into two halves of 14 participants. The fMRI signals were averaged within each half, and voxel-wise ISC was computed between the two half-group average time series. The voxel-wise ISC values were then averaged across voxels and runs to obtain one scalar ISC score for that split. Finally, the 10 split scores were averaged to obtain a stable ISC estimate for the candidate group.

This repeated split-half procedure was used because the ISC of a group should not depend on one arbitrary split. A single split can give a noisy estimate of group reliability. Repeating the split several times gives a more stable score and makes the group selection more robust.

2.5.2 Matching criterion

The selected English and Chinese groups were chosen to match the fixed French group in whole-brain ISC. For each candidate group, the stable ISC score was defined as the average of its repeated split-half estimates:

$$ISC(G) = \frac{1}{S} \sum_{s=1}^{S} ISC_s(G), \tag{2.3}$$

where $S = 10$ is the number of repeated splits.

Then select the English–Chinese pair that minimized the range of the three language scores:

$$\Delta = \max(ISC_{EN}, ISC_{CN}, ISC_{FR}) - \min(ISC_{EN}, ISC_{CN}, ISC_{FR}). \tag{2.4}$$

This criterion directly follows the purpose of the procedure: the three average subjects should have similar whole-brain reliability.

Language	**Mean**	Median	Max
English	**0.2983**	0.2711	0.7778
Chinese	**0.2988**	0.2663	0.8440
French	**0.2991**	0.2752	0.8088

Table 2.1: **Summary statistics of voxel-wise ISC values for the final matched groups.** The table reports the mean, median, and maximum voxel-wise ISC values for the selected English, Chinese, and French groups.

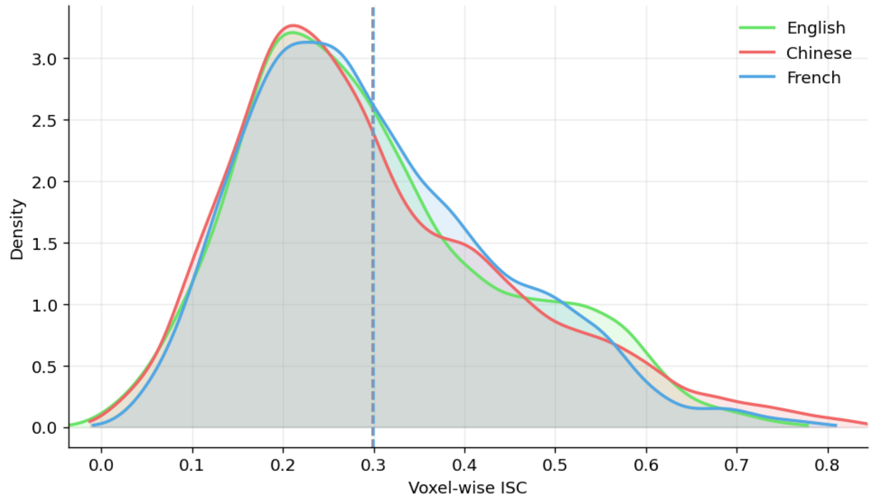

Figure 2.4: **Voxel-wise ISC distributions of the final matched groups.** The final English, Chinese, and French groups contain 28 participants each and have nearly identical scalar ISC values.

2.6 Construction of the matched average subjects

The final selected groups had highly similar reliability profiles. At the scalar level, the selected English, Chinese, and French groups were matched to have nearly identical whole-brain ISC scores. To further inspect whether this scalar matching was also reflected at the voxel level, we examined the distribution of voxel-wise ISC values in the selected groups. As shown in Table 2.1 and Figure 2.4, the three languages showed very similar distributions. The mean voxel-wise ISC values were 0.2983 for English, 0.2988 for Chinese, and 0.2991 for French. The medians were also close across languages: 0.2711, 0.2663, and 0.2752, respectively. A glass-brain visualization of the voxel-wise inter-subject reliability maps for the three selected language groups is provided in Appendix B. In these maps, color intensity represents voxel-wise ISC values, and dotted contours mark the top 25% most reliable voxels within each language.

This result shows that the three selected groups are well matched in their overall reliability. Importantly, the similarity is not limited to a single scalar score: the voxel-wise ISC distributions also largely overlap across languages. This suggests that the matched groups provide comparable fMRI targets for the following encoding analyses. The selected participant IDs for the three languages are reported in Appendix C.

After selecting the matched groups, we rebuilt the average subjects from the selected

participant lists. For each language and each run, the selected participants' fMRI time series were transformed with the common mask. The average-subject script applied detrending, standardization, and high-pass filtering with a $1/128$ Hz cut-off.
For each language and each run, the average subject was computed as the voxel-wise mean of the 28 selected participants:

$$\bar{\mathbf{Y}}_{l,k} = \frac{1}{|\mathcal{S}_l|} \sum_{i \in \mathcal{S}_l} \mathbf{Y}_{i,l,k}, \tag{2.5}$$

The resulting average time series was standardized again across time for each voxel and saved as a compressed matrix.

2.7 Summary

This chapter introduced the ISC and average-subject pipeline used in this thesis. The exploratory analyses showed that group-wise ISC increases with sample size and is affected by listener proficiency. These results confirmed that the reliability of an average brain depends on the participants included in the group. The final ISC-matching procedure then constructed comparable average subjects for English, Chinese, and French. French was fixed at 28 participants, while English and Chinese were selected from larger pools to match the French whole-brain ISC. The matched participant lists were used to rebuild one average subject per language, with 9 run-level files for each language.
The main point is methodological: before comparing model–brain alignment across languages, the brain targets must be made comparable. ISC-matched average subjects provide a practical way to control participant number and whole-brain fMRI reliability, making the later cross-language encoding analyses more interpretable.

Chapter 3

Encoding Models for Multilingual Model–Brain Alignment

The previous chapter described how ISC-matched average subjects were constructed for English, French, and Chinese. In this chapter, we use these average subjects as neural targets in an encoding analysis. The aim is to test how well representations from large language models predict brain activity during naturalistic language comprehension, and whether this model–brain alignment can be compared across languages under controlled fMRI reliability.

3.1 Method

The analysis follows a simple encoding-model framework of Bonnasse-Gahot and Pallier (2024). For each language, participants listened to *Le Petit Prince* in their native language, while the corresponding text was given to a language model. As shown in Figure 3.1, we extracted the hidden activations of the model layer by layer, aligned these activations with the word timing of the audiobook, convolved them with a canonical haemodynamic response function, and used them to predict the average-subject BOLD signal with ridge regression.

The fMRI targets were the average subjects described in the previous chapter. Each language contained 9 runs, and each run was represented as a time-by-voxel matrix:

$$Y_k \in \mathbb{R}^{T_k \times V},$$

where T_k is the number of fMRI time points in run k, and V is the number of voxels in the all-language common mask. The same common mask was used for English, French, and Chinese, so that all brain scores were computed in the same voxel space. For each run, the first and last 10 TRs were removed, corresponding to the first and last 20 seconds of data. The remaining time series were then standardized before model fitting.

We used three pretrained decoder-only language models: GPT-2 XL, Llama-3.1-8B, and Qwen3-8B from huggingface. GPT-2 XL was included as an English-dominant baseline, while Llama-3.1-8B and Qwen3-8B were used as larger recent models with stronger multilingual capacity. The models were used only as fixed feature extractors.

For each model and language, we used precomputed layer-wise activations from the corresponding text of *Le Petit Prince*. Each word in the story was associated with its onset

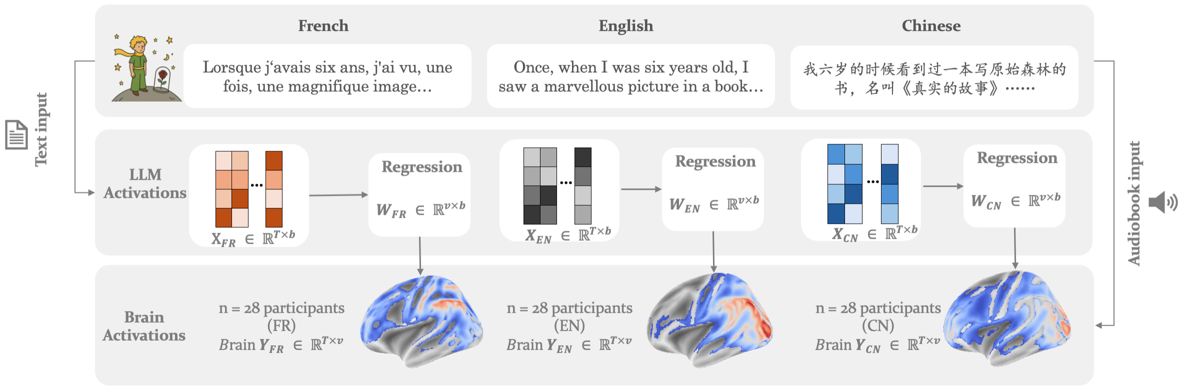

Figure 3.1: **Schematic of the multilingual encoding analysis.** For French, English, and Chinese, the corresponding text of *Le Petit Prince* was fed into a large language model to extract layer-wise activations. These were aligned with audiobook timing, convolved with a haemodynamic response function, and used in ridge regression to predict the average-subject fMRI responses. Each language was modeled separately using ISC-matched average subjects with 28 participants.

and offset time in the audiobook. For a given model layer l and run k, the model activations can be written as:

$$A_{l,k} \in \mathbb{R}^{N_k \times D_l},$$

where N_k is the number of words in run k, and D_l is the hidden-state dimension of layer l. Because model activations are defined at word times whereas fMRI signals are sampled every TR, we transformed the word-level activations into fMRI regressors.For each model feature, we created an event sequence using word onsets, word durations, and activation values. This event sequence was convolved with the Glover haemodynamic response function and sampled at the fMRI frame times. This convolution models the delayed and temporally smoothed BOLD response, so that word-level model activations are transformed into predictors at the fMRI time scale. Following previous work(Bonnasse-Gahot & Pallier, 2024), we used HRF convolution as a compact alternative to fitting multiple time-delayed copies of each feature.
This produced, for each layer and run, a design matrix:

$$X_{l,k} \in \mathbb{R}^{T_k \times D_l}.$$

The same trimming and standardization steps were then applied to the model regressors: the first and last 10 TRs were removed, and the regressors were standardized run by run. This gave one aligned pair $(X_{l,k}, Y_k)$ for each model layer and each run.
For each layer, we fitted a linear mapping from model activations to brain responses using ridge regression. The model predicts the BOLD response from the HRF-convolved model features:

$$Y = X_l W_l + \epsilon,$$

where X_l is the model-derived design matrix for layer l, Y is the fMRI response matrix, W_l is the weight matrix mapping model features to voxels, and ϵ is the residual error. The weights were estimated by minimizing:

$$\hat{W}_l = \arg\min_{W} \|Y - X_l W\|_2^2 + \alpha \|W\|_2^2 .$$

Ridge regression was used because model activations are high-dimensional and correlated. The regularization parameter α controls the strength of the penalty and helps reduce overfitting.

Model performance was evaluated with leave-one-run-out cross-validation. In each fold, one of the 9 runs was held out as the test run, and the remaining 8 runs were used for training and validation. Within these 8 runs, one run was used as validation data to select the ridge parameter α. We tested 16 logarithmically spaced values between 10^2 and 10^7. For each value of α, the model was fitted on the training runs and evaluated on the validation run. The value that gave the highest mean Pearson correlation across voxels was selected. After selecting α, the model was refitted on the 8 non-test runs and evaluated on the held-out test run. Prediction accuracy was measured separately for each voxel as the Pearson correlation between the predicted and observed BOLD time series. This procedure was repeated until each run had served once as the test run. The final brain score for layer l and voxel v was obtained by averaging the correlation values across the 9 test folds:

$$r_{l,v} = \frac{1}{9} \sum_{k=1}^{9} \mathrm{corr}\left(Y_{k,v}, \hat{Y}_{l,k,v}\right) .$$

The whole-brain score of a layer was then computed by averaging $r_{l,v}$ across all voxels in the common mask. A higher score means that the model layer provides a better linear prediction of the brain response. For each model, language, and layer, we saved three outputs: the voxel-wise correlation map, the fitted ridge coefficients, and the selected regularization parameter.

3.2 Encoding results

3.2.1 Within-language model–brain alignment

We first evaluated how well each model predicted brain activity within the same language. For each model and language, we selected the layer with the highest whole-brain prediction score. The results are shown in Table 3.1.

The within-language results show two main patterns. First, GPT-2 XL reached its highest score in English, with a brain score of 0.210, but performed lower in French and Chinese. This pattern is expected for an English-dominant model and suggests that its representations are more directly aligned with English brain responses than with non-English inputs.

Second, Llama 3.1 8B and Qwen3 8B showed more balanced within-language performance across the three languages. For Llama 3.1 8B, the best scores were 0.231 in Chinese, 0.223 in English, and 0.220 in French. For Qwen3 8B, the corresponding scores were 0.231, 0.226, and 0.220. The differences across languages were therefore relatively small for the

Model	Source Language	Best layer	Brain score
GPT-2 XL	Chinese	23	0.142
GPT-2 XL	English	25	0.210
GPT-2 XL	French	26	0.160
Llama 3.1 8B	Chinese	12	0.231
Llama 3.1 8B	English	13	0.223
Llama 3.1 8B	French	13	0.220
Qwen3 8B	Chinese	20	0.231
Qwen3 8B	English	20	0.226
Qwen3 8B	French	20	0.220

Table 3.1: **Best within-language brain scores.** For each model and language, the table reports the layer with the highest mean whole-brain prediction score.

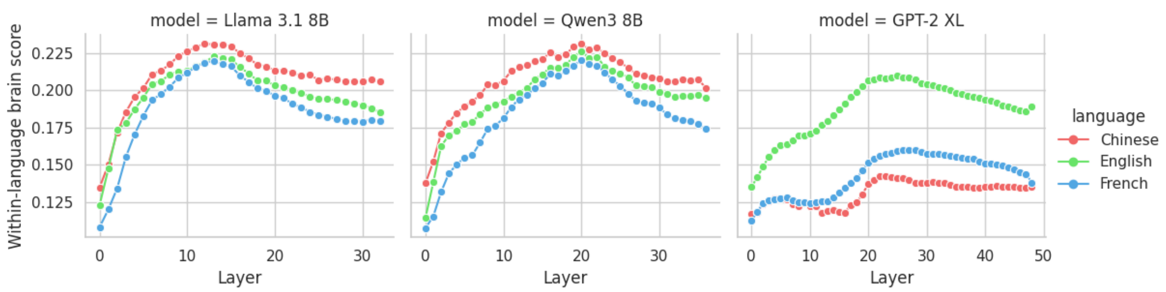

Figure 3.2: **Layer-wise within-language brain score.** The x-axis indicates model layer and the y-axis indicates the mean whole-brain prediction score. Separate curves show English, French, and Chinese. (a) GPT-2 XL shows a stronger English advantage, whereas (b) Llama 3.1 8B and (c) Qwen3 8B show more similar peak scores across languages.

two larger models. This suggests that their representations provide a more stable basis for predicting brain responses across the three languages, although this should not be interpreted as evidence that their representations are fully language-independent.

The best layers appear at the middle to late layers. GPT-2 XL peaked in around layers 23–26. Llama 3.1 8B peaked around layers 12–13, and Qwen3 8B peaked around layer 20. These layer positions are not directly comparable across models because the models have different depths and architectures. However, the results are consistent with previous model–brain alignment studies showing that intermediate-to-late representations often provide stronger predictions of brain responses during natural language comprehension Caucheteux and King (2022).

3.2.2 Cross-language transfer

3.2.2.1 Whole brain transfer patterns

We then tested whether encoding weights learned in one language could predict brain activity in another language. For each model, source language, and layer, the ridge-regression weights were trained on the source language and then applied to

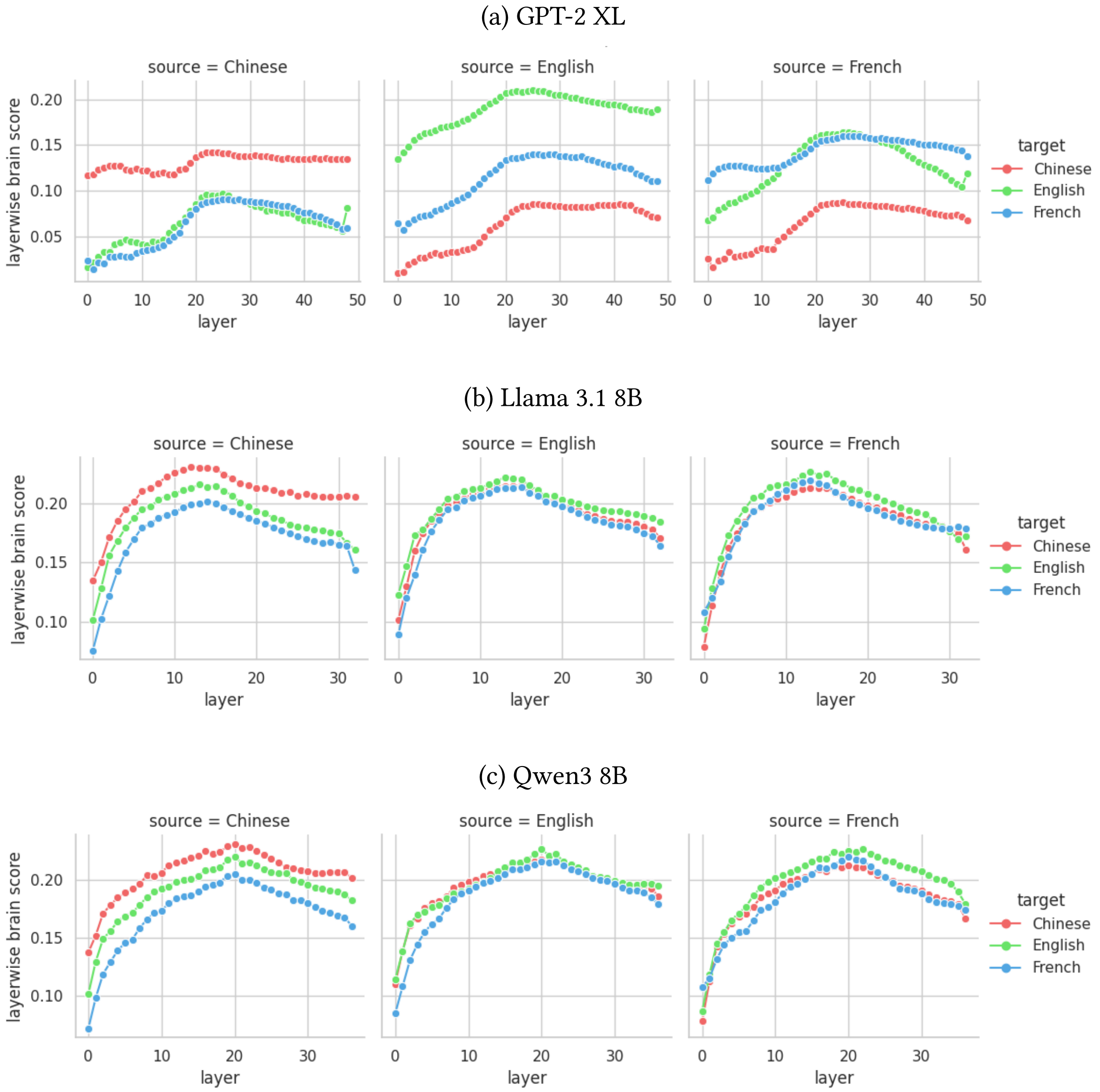

Figure 3.3: **Layer-wise within-language and cross-language transfer.** Rows show different models, and columns show the source language used to train the encoding weights. In each panel, separate curves show the target language on which the weights were evaluated. The diagonal source–target conditions correspond to within-language prediction, whereas the off-diagonal conditions correspond to cross-language transfer without re-estimating the regression weights. GPT-2 XL shows stronger language dependence, especially for conditions involving Chinese, whereas Llama 3.1 8B and Qwen3 8B show more similar layer-wise profiles across target languages.

target-language fMRI responses without re-estimating the weights. In this setting, diagonal conditions correspond to within-language prediction, whereas off-diagonal conditions correspond to strict cross-language transfer.

Figure 3.3 shows the layer-wise transfer profiles. Each row corresponds to one model, each column corresponds to one source language, and each curve corresponds to one target

Model	Source → Target	English	French	Chinese
GPT-2 XL	English	0.210	0.140	0.086
	French	0.164	0.160	0.086
	Chinese	0.096	0.089	0.142
Llama 3.1 8B	English	0.223	0.214	0.215
	French	0.228	0.220	0.214
	Chinese	0.213	0.199	0.231
Qwen3 8B	English	0.226	0.216	0.218
	French	0.225	0.220	0.212
	Chinese	0.220	0.206	0.231

Table 3.2: **Within-language and cross-language transfer matrices.** Rows indicate the source language used to train the encoding weights, and columns indicate the target language of the tested fMRI responses. Diagonal values correspond to within-language prediction. Off-diagonal values correspond to strict cross-language transfer without re-estimating the regression weights.

language.

The layer-wise transfer curves show that cross-language prediction is possible for all three models, but the degree of transfer differs strongly across models. For GPT-2 XL, the target-language curves are clearly separated. When English is the source or target language, the scores are generally higher than in conditions involving Chinese. This suggests that GPT-2 XL transfer is sensitive to the source–target language pair. For Llama 3.1 8B and Qwen3 8B, the three target-language curves are much closer to one another within each source-language panel. The curves also show a similar rise-and-fall pattern across layers. This indicates that the two larger models support more stable cross-language transfer than GPT-2 XL. To summarize the transfer results numerically, Table 3.2 reports the best-layer transfer matrices for the three models.

The transfer matrices confirm the pattern observed in the layer-wise curves. For GPT-2 XL, cross-language transfer was substantially lower than within-language prediction. The strongest condition was English-to-English, with a score of 0.210. However, transfer from English to French dropped to 0.140, and transfer from English to Chinese dropped further to 0.086. Transfer conditions involving Chinese were also low when French or Chinese was used as the source language. This pattern suggests that the encoding weights learned from GPT-2 XL are more language-dependent.

In contrast, Llama 3.1 8B and Qwen3 8B showed much smaller differences between within-language and cross-language conditions. For Llama 3.1 8B, most transfer scores remained around 0.20–0.23. For Qwen3 8B, the scores were similarly stable, mostly around 0.21–0.23. This suggests that the representations of the two larger models support a more stable mapping from model activations to brain responses across languages. Table 3.3 summarizes the transfer loss for each model.

GPT-2 XL showed the largest transfer loss. Its mean score decreased from 0.171 in the within-language conditions to 0.110 in the cross-language conditions, corresponding to a drop of 0.061. The corresponding drops were much smaller for Llama 3.1 8B and Qwen3 8B, with values of 0.011 and 0.010, respectively. Thus, cross-language transfer was much

Model	Within	Transfer	Drop	EN–FR	EN–CN
GPT-2 XL	0.171	0.110	0.061	0.152	0.091
Llama 3.1 8B	0.225	0.214	0.011	0.221	0.214
Qwen3 8B	0.226	0.216	0.010	0.221	0.219

Table 3.3: **Summary of cross-language transfer.** "Within" is the average of the three diagonal conditions in Table 3.2. "Transfer" is the average of the six off-diagonal conditions. "Drop" is the difference between within-language prediction and cross-language transfer. EN–FR averages English→French and French→English. EN–CN averages English→Chinese and Chinese→English.

closer to within-language prediction for the two larger models.

The language-pair comparison gives a similar result. For GPT-2 XL, English–French transfer was higher than English–Chinese transfer, with a difference of approximately 0.061. For Llama 3.1 8B, this difference was smaller, around 0.006. For Qwen3 8B, it was almost absent, around 0.001. This suggests that GPT-2 XL is more sensitive to the source–target language pair, whereas Llama 3.1 8B and Qwen3 8B show weaker dependence on this particular language-pair contrast.

This pattern is compatible with the idea that larger multilingual models contain representations that are more shared across languages. However, the result should not be over-interpreted. Cross-language transfer may reflect shared semantic information, but it may also be influenced by tokenization, script, pretraining distribution, source-language fit, residual differences in fMRI reliability, and the shared narrative structure of the translated story. Therefore, the present result supports a moderate conclusion: cross-language transfer is possible and is stronger for the two larger multilingual models.

As an additional exploratory analysis, we also examined cross-language transfer within a set of a priori language-related regions of interest (ROIs), including temporal, inferior frontal, temporo-parietal, and precuneus regions. The full ROI-wise results are provided in Appendix D.1. Overall, the ROI-level patterns were broadly consistent with the whole-brain results: GPT-2 XL showed stronger language-pair dependence, especially for conditions involving Chinese, whereas Llama 3.1 8B and Qwen3 8B showed more stable transfer profiles across ROIs.

3.2.3 Encoding-weight similarity across languages

The cross-language transfer analysis tested whether encoding weights trained in one language could predict brain responses in another language. As a complementary analysis, we examined whether the encoding weights themselves were similar across languages. This analysis asks whether the model-to-brain mappings learned from English, French, and Chinese have similar coefficient structure.

For each model, language, and layer, ridge regression produced a weight matrix:

$$W_{m,l,g} \in \mathbb{R}^{V \times D_l},$$

where m is the model, l is the layer, g is the language, V is the number of voxels, and D_l is the number of model features at layer l. To compare the mappings across languages, I

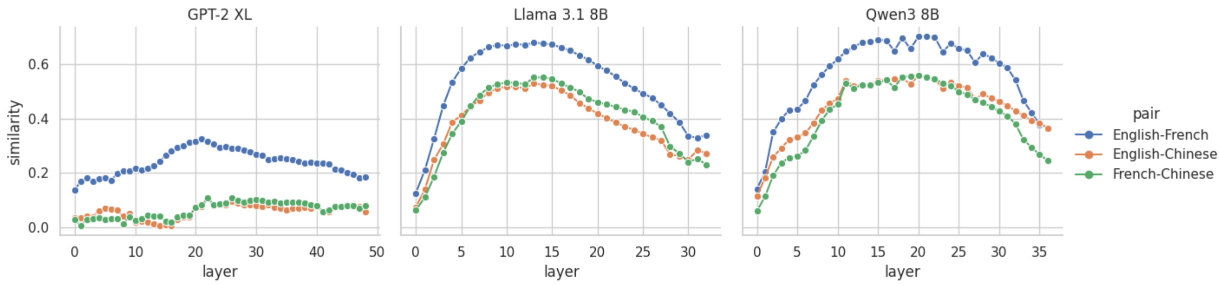

Figure 3.4: **Layer-wise beta similarity across languages.** For each model and layer, beta similarity was computed between the encoding weights learned from English, French, and Chinese. The y-axis shows Pearson correlation between beta-weight patterns, and the x-axis shows model layer. GPT-2 XL shows low beta similarity overall, especially for language pairs involving Chinese. Llama 3.1 8B and Qwen3 8B show substantially higher beta similarity, with the strongest similarity in middle layers.

Model	Focus layer	English–French	English–Chinese	French–Chinese
GPT-2 XL	26	0.290	0.097	0.108
Llama 3.1 8B	13	0.678	0.529	0.551
Qwen3 8B	20	0.700	0.558	0.559

Table 3.4: **Cross-language beta similarity at the focus layer.** For each model, beta similarity was computed at the layer used in the main encoding and transfer analyses. Values indicate Pearson correlations between language-specific beta-weight patterns. Higher values indicate more similar model-to-brain mappings across languages.

computed Pearson correlations between beta-weight patterns for each language pair: English–French, English–Chinese, and French–Chinese. Higher beta similarity indicates that two languages induce more similar model-to-brain mappings.

Figure 3.4 shows that the three models differ strongly in cross-language beta similarity. For GPT-2 XL, beta similarity remains low across layers. The English–French pair is consistently higher than the two pairs involving Chinese, but even this pair only reaches moderate values. This suggests that the model-to-brain mappings learned from GPT-2 XL are relatively language-dependent.

In contrast, Llama 3.1 8B and Qwen3 8B show much higher beta similarity across languages. For both models, beta similarity rises quickly in early layers, remains high through middle layers, and decreases toward later layers. This layer-wise profile is consistent with the idea that middle layers of multilingual models provide more shared cross-linguistic representations, whereas early and late layers may be more affected by language-specific form or output-related structure.

To summarize the result numerically, Table 3.4 reports beta similarity at the focus layer of each model. The focus layers correspond to the layers used in the main encoding and transfer analyses.

The focus-layer results mirror the cross-language transfer results. GPT-2 XL shows relatively low beta similarity across languages. Its English–French similarity is higher than English–Chinese and French–Chinese, suggesting a stronger correspondence between English and French than between either language and Chinese. However, all three values are much lower than those observed for the two larger models.

Llama 3.1 8B and Qwen3 8B show substantially higher beta similarity across all language pairs. For Llama 3.1 8B, English–French similarity reaches 0.678, while English–Chinese and French–Chinese are 0.529 and 0.551. For Qwen3 8B, English–French similarity reaches 0.700, and the two pairs involving Chinese are also clearly positive, around 0.56. This suggests that the two larger models learn more similar model-to-brain mappings across languages than GPT-2 XL.

These results provide complementary evidence for partial cross-language sharing in the multilingual models. It supports the more moderate conclusion that Llama 3.1 8B and Qwen3 8B show more similar cross-language encoding-weight structure than GPT-2 XL.

Chapter 4

Discussion

4.1 Summary of the study

This thesis investigated whether multilingual large language models can provide a shared representational basis for predicting brain activity during naturalistic language comprehension across English, French, and Chinese. The central motivation was to test whether model–brain alignment is tied to language-specific surface form, or whether part of the alignment reflects representations that are shared across languages.

A first methodological contribution of the thesis was the construction of ISC-matched average subjects. Because the three language groups in the *Le Petit Prince* fMRI dataset had different numbers of participants, directly averaging all available participants in each language would have produced fMRI targets with different levels of reliability. To reduce this confound, English and Chinese participant groups were selected to match the fixed French group in both participant number and whole-brain inter-subject correlation. This procedure made the subsequent model–brain comparisons more interpretable, because differences in brain score were less likely to be explained only by differences in the reliability of the average fMRI target.

A second contribution was the comparison of within-language and cross-language encoding models across three language models: GPT-2 XL, Llama 3.1 8B, and Qwen3 8B. For each model and language, layer-wise activations were aligned with the audiobook timing, convolved with a canonical haemodynamic response function, and used in ridge-regression encoding models to predict the average-subject fMRI responses. This made it possible to compare not only how well each model predicted brain activity within a language, but also whether encoding weights learned in one language could transfer to another language without being re-estimated.

Finally, the thesis complemented the transfer analysis with beta-pattern similarity analyses. These analyses compared the structure of the encoding weights learned from different languages. The goal was to test whether successful transfer was accompanied by similar model-to-brain mappings across English, French, and Chinese. Additional ROI-wise analyses were included as exploratory results to examine whether the whole-brain patterns were also visible in a priori language-related and narrative-related regions.

4.2 Implications for the research questions

The first research question was whether model–brain alignment can transfer across languages. The answer is yes, but with important qualifications. Cross-language transfer was possible for all three models, but it was much stronger and more stable for Llama 3.1 8B and Qwen3 8B than for GPT-2 XL. This suggests that at least part of the mapping between language-model representations and brain responses is not language-specific. However, the result should not be interpreted as showing that all languages are represented identically, either in the models or in the brain.

The second question was whether multilingual models provide a more balanced basis for brain prediction than an English-dominant model. The results support this interpretation. GPT-2 XL showed a clear English advantage and a larger drop in cross-language transfer. Llama 3.1 8B and Qwen3 8B showed more similar within-language scores across the three languages, smaller transfer losses, and higher cross-language beta similarity. These converging observations suggest that the larger multilingual models encode representations that are more reusable across languages.

The third question was whether the results support the existence of a shared semantic hub. The evidence is consistent with this possibility, but it does not prove it in a strong mechanistic sense. The strongest support comes from the combination of three observations: cross-language transfer remained possible under a strict no-retraining setting; transfer loss was small for the two multilingual models; and encoding-weight patterns were more similar across languages in these models, especially in middle layers. Together, these findings are compatible with the hypothesis that multilingual LLMs and the human brain share access to language-general semantic or conceptual representations during story comprehension.

However, the term "shared semantic hub" should be understood as a working interpretation rather than a demonstrated anatomical or mechanistic entity. The present analyses do not isolate a single brain region, a single model component, or a single set of neurons that implements such a hub. They show that model-derived representations can support cross-language prediction of brain activity, and that this transfer is stronger in models expected to have richer multilingual representations. This supports a moderate conclusion: the results are consistent with partially shared semantic representations across languages, while also leaving room for language-specific processing.

4.3 Limitations

Several limitations should be considered when interpreting these results.

First, the three languages differ in many ways beyond semantic content. English, French, and Chinese differ in script, phonology, morphology, word segmentation, tokenization, audiobook timing, and translation choices. Cross-language transfer may therefore reflect a mixture of shared semantic structure, shared narrative structure, model-specific tokenization effects, and language-specific processing demands. The present design cannot fully separate these factors.

Second, model comparisons are confounded by differences in model size, training data, architecture, tokenizer, and multilingual exposure. GPT-2 XL, Llama 3.1 8B, and Qwen3

8B are not controlled variants of the same model family. Therefore, the stronger transfer observed in Llama 3.1 8B and Qwen3 8B should not be attributed only to multilinguality. It may also reflect larger scale, more recent training data, better contextual representations, or different tokenization strategies.

Finally, the ROI-wise analyses remain exploratory. They are useful for checking whether whole-brain patterns are also visible in language-related and narrative-related regions, but they are not sufficient to establish a clear brain-region interaction. Some ROI-level transfer scores were close across conditions, and some off-diagonal transfer conditions even slightly exceeded within-language conditions. These effects should be interpreted cautiously, because they may reflect noise, averaging, or differences in source-language fit rather than genuine superiority of cross-language prediction.

4.4 Future work

Future work could extend this project in several directions.

A first direction is to expand the model comparison. The present study compared one English-dominant model and two larger multilingual models. A stronger design would compare models that vary systematically in multilingual exposure while holding size and architecture more constant. It would also be useful to include additional baselines, such as random vectors, random embeddings, static word embeddings, and untrained language models. These baselines would help separate the contribution of lexical identity, distributional semantics, contextualization, and large-scale pretraining.

A second direction is to connect the encoding results more directly to model interpretability. The present results suggest that middle layers of multilingual models may support stronger cross-language transfer and beta similarity. Future work could test whether this effect is driven by specific multilingual features, language-specific neurons, or sparse-autoencoder features. Metrics such as language activation probability entropy, neuron-level language selectivity, or SAE-based feature selectivity could be compared with brain-alignment scores. This would make it possible to ask whether the model components that support cross-language transfer in NLP tasks are also the components that best predict cross-language brain responses.

A third direction is to separate semantic, syntactic, and low-level linguistic contributions. The present analysis uses full LLM activations, which combine many types of information. Future work could use controlled linguistic probes, minimal pairs, or feature ablation to ask whether cross-language brain transfer is mainly supported by semantic representations, syntactic representations, or lower-level lexical and acoustic features. This would be especially useful for testing the shared-semantic-hub interpretation more directly.

A forth direction is to deepen the brain-region analysis. The exploratory ROI results suggest that high-level temporal, inferior frontal, temporo-parietal, and precuneus regions may show relatively stable transfer for the multilingual models. However, the current analysis does not yet provide a clear interaction between model, language pair, layer, and brain region. Future work could use more systematic ROI definitions, voxel-wise statistical maps, and mixed-effects models to test whether cross-language transfer is stronger in semantic and narrative regions than in lower-level auditory or phonological regions.

4.5 Conclusion

This thesis provides a controlled analysis of multilingual model–brain alignment using ISC-matched average subjects in the trilingual *Le Petit Prince* fMRI dataset. The results show that cross-language transfer of encoding models is possible, especially for Llama 3.1 8B and Qwen3 8B. Compared with GPT-2 XL, these two models showed more balanced within-language brain scores, smaller cross-language transfer losses, and higher beta-pattern similarity across languages.

These findings support a moderate account of multilingual model–brain alignment. They suggest that part of the alignment between LLM representations and brain activity reflects language-shared representations, likely related to semantic or narrative-level comprehension. At the same time, the results do not imply that language-specific processing disappears. Transfer remained model-dependent, language-pair differences were still visible in some analyses, and the precise neural and computational mechanisms remain to be identified. The strongest conclusion is therefore not that a fully language-independent semantic hub has been proven, but that the present results are consistent with a partially shared representational space linking multilingual LLMs and human brain responses during naturalistic story comprehension.

Bibliography

Bhattacharya, S., & Bojar, O. (2023). Unveiling multilinguality in transformer models: Exploring language specificity in feed-forward networks. *arXiv preprint arXiv:2310.15552*.

Bonnasse-Gahot, L., & Pallier, C. (2024). fmri predictors based on language models of increasing complexity recover brain left lateralization. In *Neurips 2024-38th conference on neural information processing systems*.

Caucheteux, C., & King, J.-R. (2022). Brains and algorithms partially converge in natural language processing. *Communications biology*, *5*(1), 134.

Chen, C., Gong, X. L., Tseng, C., Klein, D. L., Gallant, J. L., & Deniz, F. (2026). Bilingual language processing relies on shared semantic representations that are modulated by each language. *Proceedings of the National Academy of Sciences*, *123*(9), e2503721123.

Chi, E. A., Hewitt, J., & Manning, C. D. (2020). Finding universal grammatical relations in multilingual bert. In *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 5564–5577).

Choenni, R., Garrette, D., & Shutova, E. (2023). Cross-lingual transfer with language-specific subnetworks for low-resource dependency parsing. *Computational Linguistics*, *49*(3), 613–641.

Choenni, R., Shutova, E., & Garrette, D. (2024). Examining modularity in multilingual lms via language-specialized subnetworks. In *Findings of the association for computational linguistics: Naacl 2024* (pp. 287–301).

Conneau, A., Wu, S., Li, H., Zettlemoyer, L., & Stoyanov, V. (2020). Emerging cross-lingual structure in pretrained language models. In *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 6022–6034).

Correia, J., Formisano, E., Valente, G., Hausfeld, L., Jansma, B., & Bonte, M. (2014). Brain-based translation: fmri decoding of spoken words in bilinguals reveals language-independent semantic representations in anterior temporal lobe. *Journal of Neuroscience*, *34*(1), 332–338.

Deng, B., Wan, Y., Yang, B., Zhang, Y., & Feng, F. (2025). Unveiling language-specific features in large language models via sparse autoencoders. In *Proceedings of the 63rd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 4563–4608).

de Varda, A. G., Malik-Moraleda, S., Tuckute, G., & Fedorenko, E. (2025). Multilingual computational models reveal shared brain responses to 21 languages. *bioRxiv*, 2025–02.

Dumas, C., Wendler, C., Veselovsky, V., Monea, G., & West, R. (2025). Separating tongue from thought: Activation patching reveals language-agnostic concept representations in transformers. In *Proceedings of the 63rd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 31822–31841).

Ferrando, J., & Costa-Jussà, M. R. (2024). On the similarity of circuits across languages: a case study on the subject-verb agreement task. In *Findings of the association for computational linguistics: Emnlp 2024* (pp. 10115–10125).

Gu, C., Nastase, S. A., Zada, Z., & Li, P. (2025). Reading comprehension in l1 and l2 readers: neurocomputational mechanisms revealed through large language models. *npj Science of Learning*, *10*(1), 46.

Hertrich, I., Dietrich, S., & Ackermann, H. (2020). The margins of the language network in the brain. *Frontiers in Communication*, *5*, 519955.

Honey, C. J., Thompson, C. R., Lerner, Y., & Hasson, U. (2012). Not lost in translation: neural responses shared across languages. *Journal of Neuroscience*, *32*(44), 15277–15283.

Hou, Z., Li, H., Gao, L., Ou, J., & Xu, M. (2024). Differential neural representations of syntactic and semantic information across languages in chinese-english bilinguals. *NeuroImage*, *303*, 120928.

Kojima, T., Okimura, I., Iwasawa, Y., Yanaka, H., & Matsuo, Y. (2024). On the multilingual ability of decoder-based pre-trained language models: Finding and controlling language-specific neurons. In *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies (volume 1: Long papers)* (pp. 6919–6971).

Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis-connecting the branches of systems neuroscience. *Frontiers in systems neuroscience*, *2*, 249.

Libovickỳ, J., Rosa, R., & Fraser, A. (2020). On the language neutrality of pre-trained multilingual representations. In *Findings of the association for computational linguistics: Emnlp 2020* (pp. 1663–1674).

Lin, N., Zhang, X., Wang, X., & Wang, S. (2024). The organization of the semantic network as reflected by the neural correlates of six semantic dimensions. *Brain and Language*, *250*, 105388.

Lindsey, J., Gurnee, W., Ameisen, E., Chen, B., Pearce, A., Turner, N. L., … Batson, J. (2025). On the biology of a large language model. *Transformer Circuits Thread*.

Liu, W., Xu, Y., Xu, H., Chen, J., Hu, X., & Wu, J. (2024). Unraveling babel: Exploring multilingual activation patterns of llms and their applications. In *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 11855–11881).

Malik-Moraleda, S., Ayyash, D., Gallée, J., Affourtit, J., Hoffmann, M., Mineroff, Z., … Fedorenko, E. (2022a). An investigation across 45 languages and 12 language families reveals a universal language network. *Nature neuroscience*, *25*(8), 1014–1019.

Malik-Moraleda, S., Ayyash, D., Gallée, J., Affourtit, J., Hoffmann, M., Mineroff, Z., … Fedorenko, E. (2022b). An investigation across 45 languages and 12 language families reveals a universal language network. *Nature neuroscience*, *25*(8), 1014–1019.

Mitchell, T. M., Shinkareva, S. V., Carlson, A., Chang, K.-M., Malave, V. L., Mason, R. A., & Just, M. A. (2008). Predicting human brain activity associated with the meanings of nouns. *science*, *320*(5880), 1191–1195.

Naselaris, T., Kay, K. N., Nishimoto, S., & Gallant, J. L. (2011). Encoding and decoding in fmri. *Neuroimage*, *56*(2), 400–410.

Nieuwland, M. S., Martin, A. E., & Carreiras, M. (2012). Brain regions that process case: evidence from basque. *Human brain mapping*, *33*(11), 2509–2520.

Oota, S., Gupta, M., & Toneva, M. (2023). Joint processing of linguistic properties in brains and language models. *Advances in Neural Information Processing Systems*, *36*, 18001–18014.

Pallier, C., Devauchelle, A.-D., & Dehaene, S. (2011). Cortical representation of the constituent structure of sentences. *Proceedings of the National Academy of Sciences*, *108*(6), 2522–2527.

Pires, T., Schlinger, E., & Garrette, D. (2019). How multilingual is multilingual bert? In *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 4996–5001).

Price, A. R., Bonner, M. F., Peelle, J. E., & Grossman, M. (2015). Converging evidence for the neuroanatomic basis of combinatorial semantics in the angular gyrus. *Journal of Neuroscience*, *35*(7), 3276–3284.

Price, C. J. (2012). A review and synthesis of the first 20 years of pet and fmri studies of heard speech, spoken language and reading. *Neuroimage*, *62*(2), 816–847.

Qin, L., Chen, Q., Zhou, Y., Chen, Z., Li, Y., Liao, L., … Yu, P. S. (2025). A survey of multilingual large language models. *Patterns*, *6*(1).

Tang, T., Luo, W., Huang, H., Zhang, D., Wang, X., Zhao, W. X., … Wen, J.-R. (2024). Language-specific neurons: The key to multilingual capabilities in large language models. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 5701–5715).

Vigneau, M., Beaucousin, V., Hervé, P.-Y., Duffau, H., Crivello, F., Houde, O., ... Tzourio-Mazoyer, N. (2006). Meta-analyzing left hemisphere language areas: phonology, semantics, and sentence processing. *Neuroimage*, *30*(4), 1414–1432.

Wei, X., Adamson, H., Schwendemann, M., Goucha, T., Friederici, A. D., & Anwander, A. (2023). Native language differences in the structural connectome of the human brain. *Neuroimage*, *270*, 119955.

Wendler, C., Veselovsky, V., Monea, G., & West, R. (2024). Do llamas work in english? on the latent language of multilingual transformers. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 15366–15394).

Wu, S., & Dredze, M. (2019). Beto, bentz, becas: The surprising cross-lingual effectiveness of bert. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (emnlp-ijcnlp)* (pp. 833–844).

Wu, Z., Yu, X. V., Yogatama, D., Lu, J., & Kim, Y. (2024). The semantic hub hypothesis: Language models share semantic representations across languages and modalities. *arXiv preprint arXiv:2411.04986*.

Xu, M., Li, D., & Li, P. (2021). Brain decoding in multiple languages: Can cross-language brain decoding work? *Brain and language*, *215*, 104922.

Zaccarella, E., Schell, M., & Friederici, A. D. (2017). Reviewing the functional basis of the syntactic merge mechanism for language: A coordinate-based activation likelihood estimation meta-analysis. *Neuroscience & Biobehavioral Reviews*, *80*, 646–656.

Zada, Z., Nastase, S. A., Li, J., & Hasson, U. (2025). Brains and language models converge on a shared conceptual space across different languages. *arXiv preprint arXiv:2506.20489*.

Zeng, H., Han, S., Chen, L., & Yu, K. (2025). Converging to a lingua franca: Evolution of linguistic regions and semantics alignment in multilingual large language models. In *Proceedings of the 31st international conference on computational linguistics* (pp. 10602–10617).

Zhang, Z., Zhao, J., Zhang, Q., Gui, T., & Huang, X.-J. (2024). Unveiling linguistic regions in large language models. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 6228–6247).

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Appendix A

Comprehension Quiz Analysis

Half size n	Total size $2n$	Top ISC	Bottom ISC	Random ISC
4	8	0.160 ± 0.003	0.048 ± 0.005	0.109 ± 0.021
8	16	0.244 ± 0.004	0.126 ± 0.008	0.189 ± 0.025
12	24	0.309 ± 0.004	0.208 ± 0.005	0.250 ± 0.029
16	32	0.361 ± 0.004	0.260 ± 0.007	0.315 ± 0.010
20	40	0.383 ± 0.004	0.330 ± 0.007	0.350 ± 0.014
24	48	0.391 ± 0.005	0.389 ± 0.003	0.389 ± 0.006

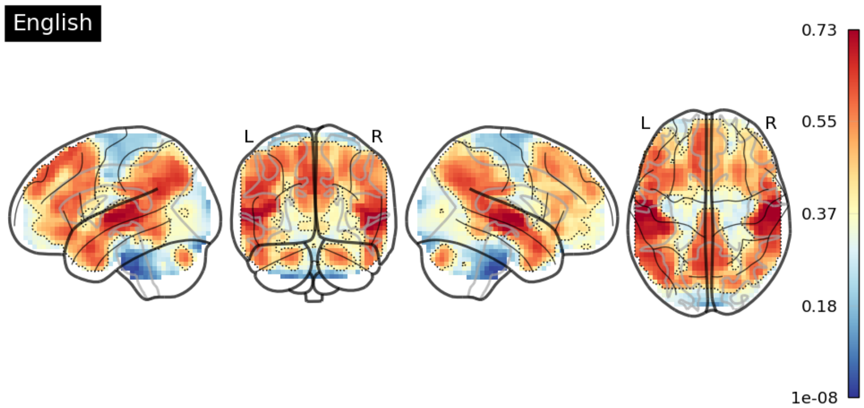
Table A.1: **English comprehension quiz analysis.** Participants were ranked by the number of correctly answered comprehension questions. Values show mean whole-brain ISC $\pm$ SD across resampling iterations.

Half size n	Total size $2n$	Top ISC	Bottom ISC	Random ISC
4	8	0.159 ± 0.004	0.106 ± 0.005	0.116 ± 0.017
8	16	0.241 ± 0.005	0.180 ± 0.004	0.211 ± 0.011
12	24	0.296 ± 0.005	0.259 ± 0.003	0.276 ± 0.011
16	32	0.344 ± 0.004	0.329 ± 0.006	0.328 ± 0.006

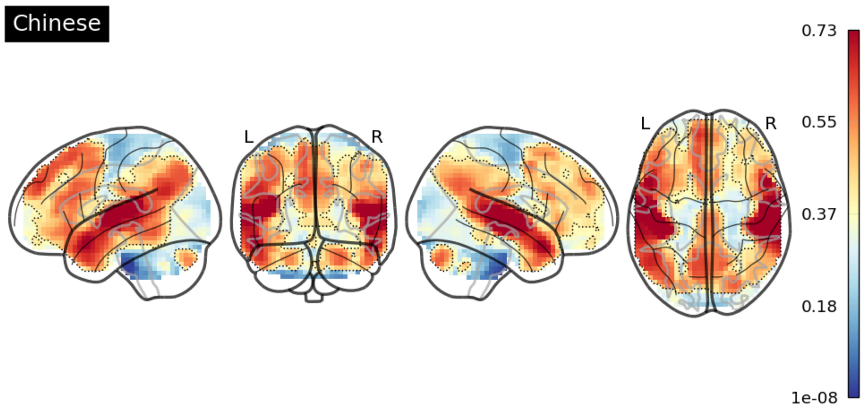
Table A.2: **Chinese comprehension quiz analysis.** Participants were ranked by the number of correctly answered comprehension questions. Values show mean whole-brain ISC $\pm$ SD across resampling iterations.

Appendix B

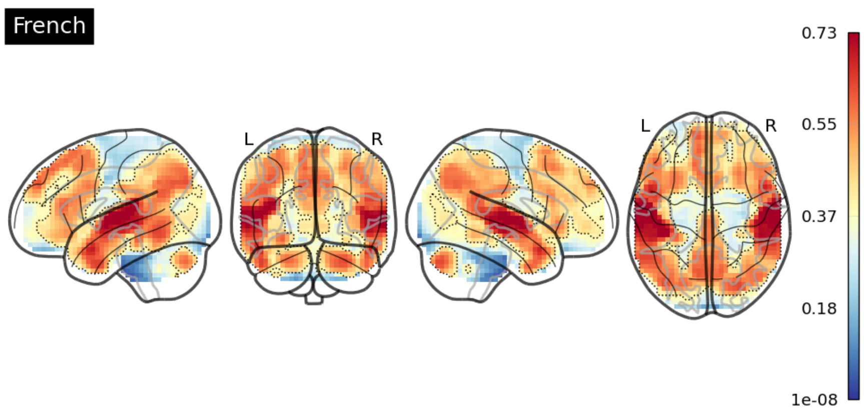
Glass-brain visualization of voxel-wise ISC reliability



(a) English



(b) Chinese



(c) French

Figure B.1: **Glass-brain representation of voxel-wise inter-subject reliability.** The three panels show voxel-wise ISC maps for the final selected English, Chinese, and French groups. Hotter colors indicate higher inter-subject reliability. Dotted contours mark the top 25% most reliable voxels within each language-specific ISC map.

Appendix C

Matched Participant IDs

C.1 English

The selected English participants were:

sub-EN057, sub-EN058, sub-EN061, sub-EN063, sub-EN064, sub-EN065, sub-EN067, sub-EN070, sub-EN072, sub-EN075, sub-EN078, sub-EN079, sub-EN081, sub-EN083, sub-EN084, sub-EN086, sub-EN087, sub-EN091, sub-EN092, sub-EN096, sub-EN097, sub-EN098, sub-EN100, sub-EN103, sub-EN105, sub-EN109, sub-EN113, sub-EN114.

C.2 Chinese

The selected Chinese participants were:

sub-CN001, sub-CN002, sub-CN003, sub-CN006, sub-CN007, sub-CN008, sub-CN009, sub-CN010, sub-CN011, sub-CN013, sub-CN014, sub-CN015, sub-CN017, sub-CN020, sub-CN021, sub-CN022, sub-CN023, sub-CN024, sub-CN025, sub-CN026, sub-CN028, sub-CN030, sub-CN031, sub-CN032, sub-CN033, sub-CN034, sub-CN036, sub-CN037.

C.3 French

The selected French participants were:

sub-FR001, sub-FR002, sub-FR003, sub-FR004, sub-FR005, sub-FR006, sub-FR007, sub-FR008, sub-FR009, sub-FR010, sub-FR011, sub-FR012, sub-FR013, sub-FR014, sub-FR015, sub-FR016, sub-FR017, sub-FR018, sub-FR019, sub-FR020, sub-FR022, sub-FR023, sub-FR024, sub-FR025, sub-FR026, sub-FR028, sub-FR029, sub-FR030.

Appendix D

ROI-wise cross-language transfer

D.1 ROI-wise cross-language transfer

As a supplementary analysis, we examined whether the cross-language transfer patterns observed at the whole-brain level were also visible in a set of a priori regions of interest (ROIs). We adopted the same set of a priori regions of interest (ROIs) as in previous studies on syntactic and semantic composition Bonnasse-Gahot and Pallier (2024), including the Temporal Pole (TP), anterior Superior Temporal Sulcus (aSTS), and posterior Superior Temporal Sulcus (pSTS) Pallier, Devauchelle, and Dehaene (2011), the Inferior Frontal Gyrus pars opercularis (BA44), triangularis (BA45), and orbitalis (BA47) Zaccarella, Schell, and Friederici (2017), and the Angular Gyrus/Temporo-Parietal Junction (AG/TPJ) A. R. Price, Bonner, Peelle, and Grossman (2015).
Following the protocol of Bonnasse-Gahot and Pallier (2024), all ROIs were defined as spheres of 10 mm radius centered on MNI152 coordinates: TP (-48, 15, -27), aSTS (-54, -12, -12), pSTS (-51, -39, 3), AG/TPJ (-52, -56, 22), BA44 (-50, 12, 16), BA45 (-52, 28, 10), BA47 (-44, 34, -8). In addition, we included the Precuneus (-4, -62, 44) as an ROI to account for its involvement in higher-level integrative processes.

ROI	**Main functional associations**
TP	Lexical-semantic integration and conceptual memory.
aSTS	Sentence-level syntactic and semantic composition.
pSTS	Speech, phonological, and auditory-linguistic processing.
BA44	Syntactic structure building and phonological/articulatory encoding.
BA45	Controlled semantic retrieval and lexical selection.
BA47	Semantic control, integration, and conceptual retrieval.
AG/TPJ	Multimodal semantic integration, discourse comprehension, attention, and social cognition.
Precuneus	Episodic memory, mental imagery, self-referential processing, and narrative-level integration.

Table D.1: Functional associations of the selected ROIs. The descriptions provide broad anatomical-functional guidance for interpreting ROI-wise model–brain alignment results.

For each model, source language, target language, and ROI, voxel-wise prediction scores were averaged within the ROI. Diagonal source–target conditions correspond to within-language prediction, whereas off-diagonal conditions correspond to cross-language transfer without re-estimating the encoding weights. Whole-brain scores are also shown as a reference.

Figure D.1, D.2, D.3 shows the ROI-wise transfer patterns for GPT-2 XL, Llama 3.1 8B, and Qwen3 8B. The ROI-level results broadly mirrored the whole-brain transfer analysis. GPT-2 XL showed stronger language-pair dependence: English–French transfer was generally better preserved than transfer involving Chinese, and Chinese-related transfer conditions were often lower across ROIs. In contrast, Llama 3.1 8B and Qwen3 8B showed more homogeneous transfer profiles, with off-diagonal cross-language conditions often close to the diagonal within-language conditions.

The three models showed different ROI-wise transfer profiles. GPT-2 XL displayed a clear language-pair dependence. In most ROIs, the English–English condition produced the highest scores, and English–French or French–English transfer was generally better preserved than transfer involving Chinese. For example, in aSTS, TP, AG/TPJ, BA45, BA47, and Precuneus, transfer between English and French remained relatively high, whereas English–Chinese and French–Chinese transfer was lower. This pattern is consistent with the whole-brain transfer results and suggests that GPT-2 XL representations support a less stable cross-language mapping, especially when the target or source language is Chinese.

In contrast, Llama 3.1 8B showed much more homogeneous ROI-wise transfer. Across most ROIs, the off-diagonal transfer conditions were close to the diagonal within-language conditions. This was especially visible in AG/TPJ, BA45, aSTS, and Precuneus, where cross-language transfer remained high for all three language pairs. Some off-diagonal conditions even slightly exceeded the corresponding within-language condition. These differences should not be over-interpreted as evidence that cross-language prediction is intrinsically better than within-language prediction; they may reflect noise, ROI-level averaging, or small differences in source-language fit. The important pattern is that transfer loss was small across ROIs.

Qwen3 8B showed a very similar pattern to Llama 3.1 8B. ROI-wise scores were high and relatively balanced across language pairs, especially in AG/TPJ, BA45, BA47, aSTS, and Precuneus. Conditions involving Chinese did not show the strong degradation observed for GPT-2 XL. This suggests that Qwen3 8B provides a more language-shared representational basis for predicting brain responses across English, French, and Chinese.

Overall, the ROI-wise analysis provides anatomically focused descriptive evidence for the main transfer result. GPT-2 XL shows stronger language-pair dependence, whereas Llama 3.1 8B and Qwen3 8B show more stable cross-language transfer across language-related ROIs. This pattern is compatible with the idea that larger multilingual models contain representations that are more shared across languages.

D.2 ROI-wise beta similarity

We also examined whether the encoding weights learned from different languages showed similar coefficient patterns within each ROI. This analysis complements the ROI-wise transfer analysis: transfer evaluates whether weights trained in one language

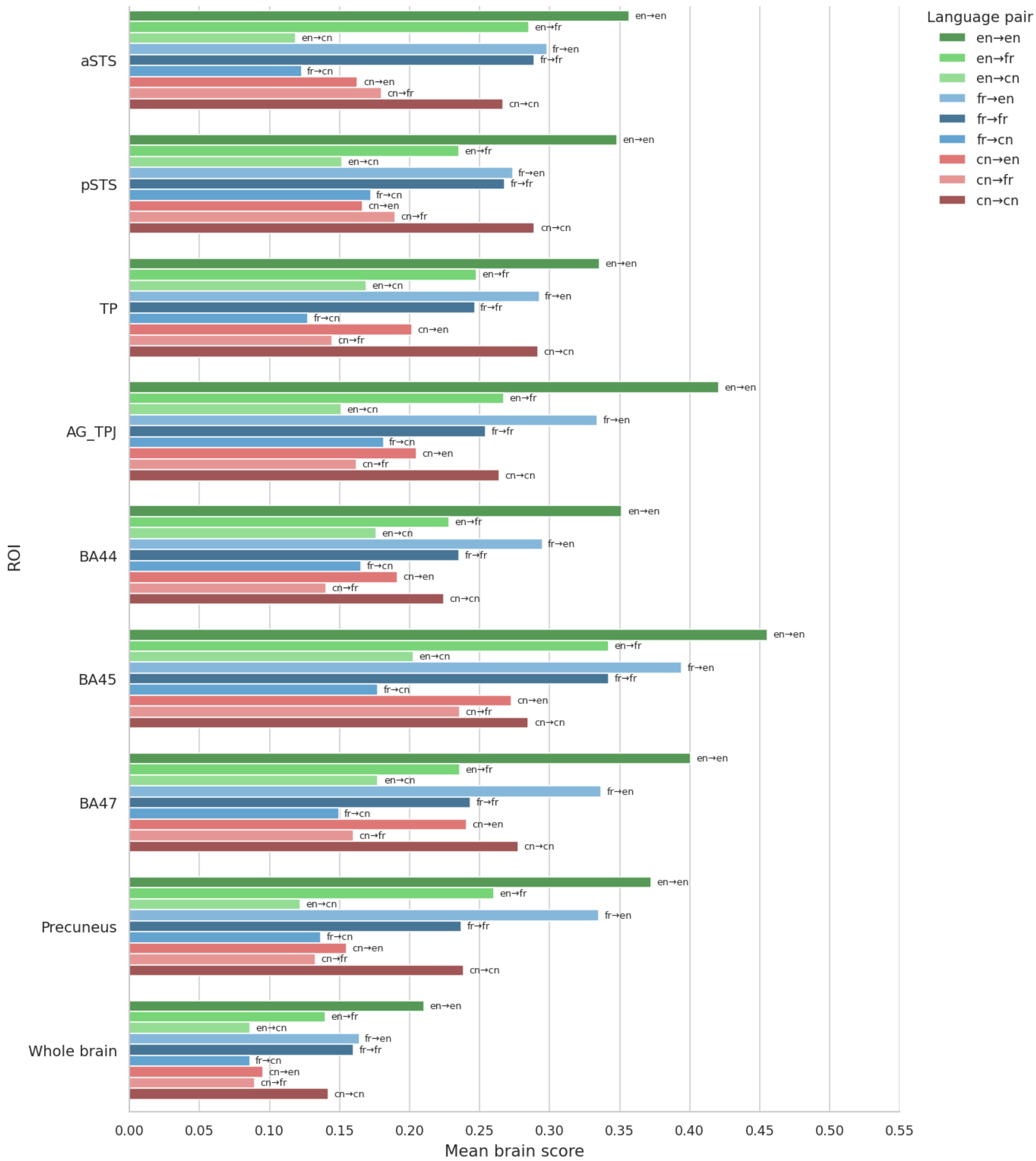


Figure D.1: ROI-wise cross-language transfer patterns for GPT-2 XL.

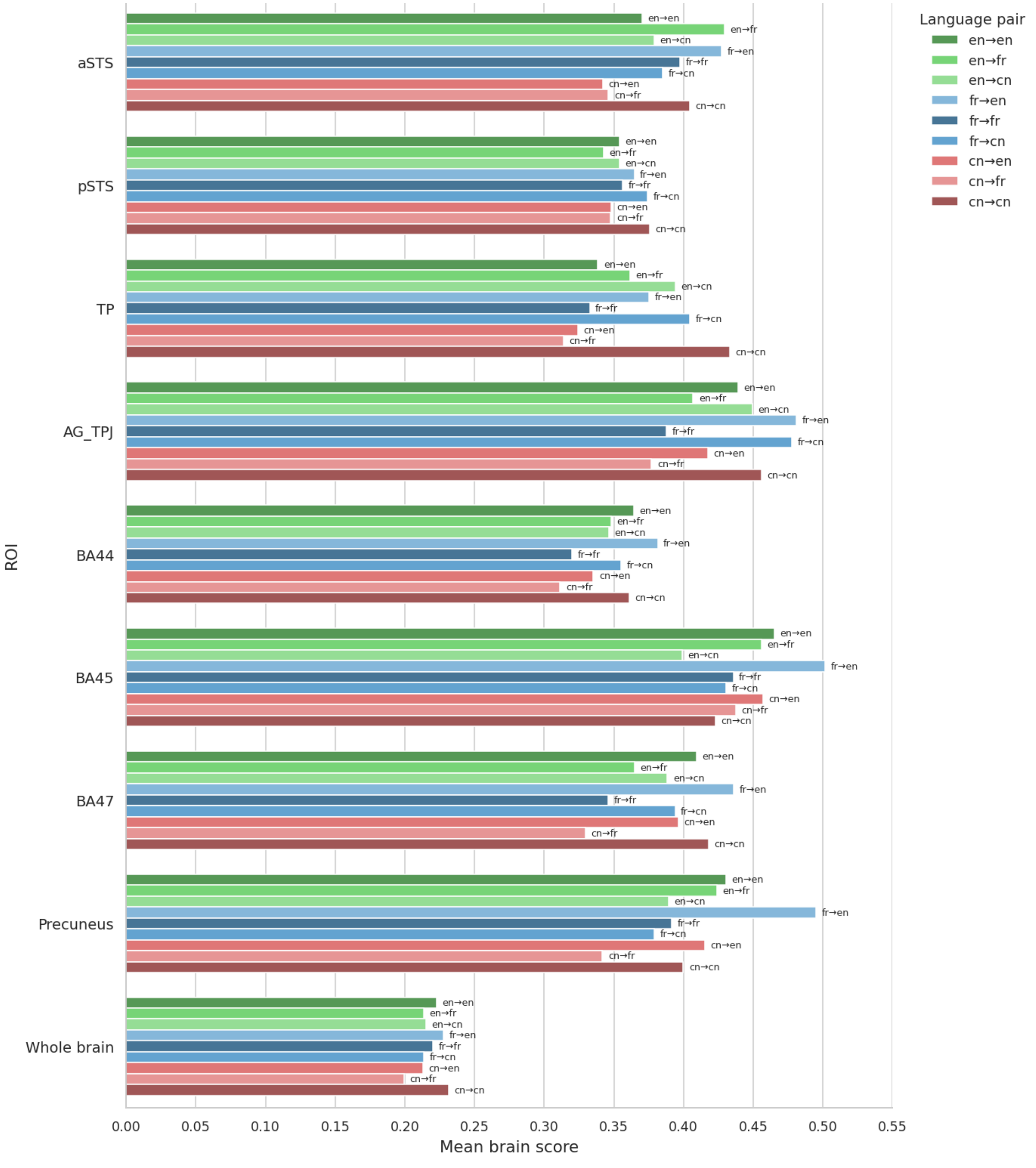


Figure D.2: ROI-wise cross-language transfer patterns for Llama 3.1 8B.

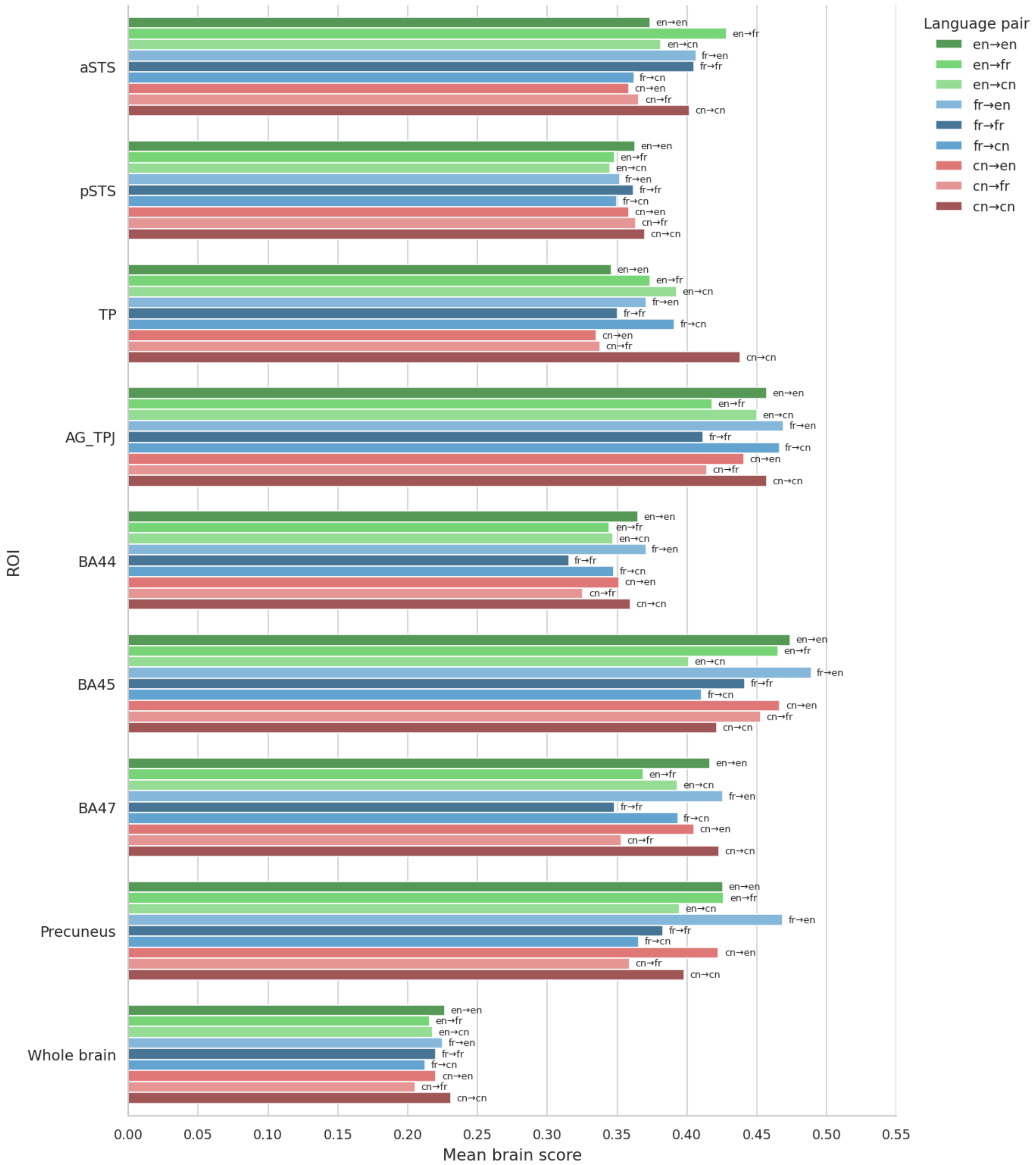


Figure D.3: ROI-wise cross-language transfer patterns for Qwen3 8B.

can predict responses in another language, whereas beta similarity directly compares the structure of the learned encoding weights across languages.

For each model, we used the same focus layer as in the main beta-similarity analysis: layer 26 for GPT-2 XL, layer 13 for Llama 3.1 8B, and layer 20 for Qwen3 8B. For each ROI, beta-weight similarity was computed for the three language pairs: English–French, English–Chinese, and French–Chinese. Higher values indicate more similar model-to-brain mappings across languages.

Figure D.4, D.5, D.6 shows the ROI-wise beta similarity results. The patterns broadly mirror the whole-brain beta-similarity analysis. GPT-2 XL showed low beta similarity overall. English–French similarity was consistently higher than the two pairs involving Chinese, whereas English–Chinese and French–Chinese similarities remained low in most ROIs. This suggests that the encoding-weight structure learned from GPT-2 XL is more language-dependent.

In contrast, Llama 3.1 8B and Qwen3 8B showed substantially higher beta similarity across ROIs. For both models, English–French similarity was generally the highest, but English–Chinese and French–Chinese similarities were also clearly positive and relatively stable across regions. This pattern was visible in temporal, inferior frontal, temporo-parietal, and precuneus ROIs. The whole-brain beta similarity was lower than ROI-level beta similarity, likely because the whole-brain average includes many voxels that are less directly involved in language comprehension.

Overall, the ROI-wise beta-similarity analysis provides descriptive support for the main result. GPT-2 XL shows weaker and more language-pair-dependent cross-language encoding-weight similarity, whereas Llama 3.1 8B and Qwen3 8B show more similar model-to-brain mappings across English, French, and Chinese. As with the ROI-wise transfer analysis, these results should be interpreted cautiously because they are based on group-average data and do not provide subject-level statistical evidence for regional differences.

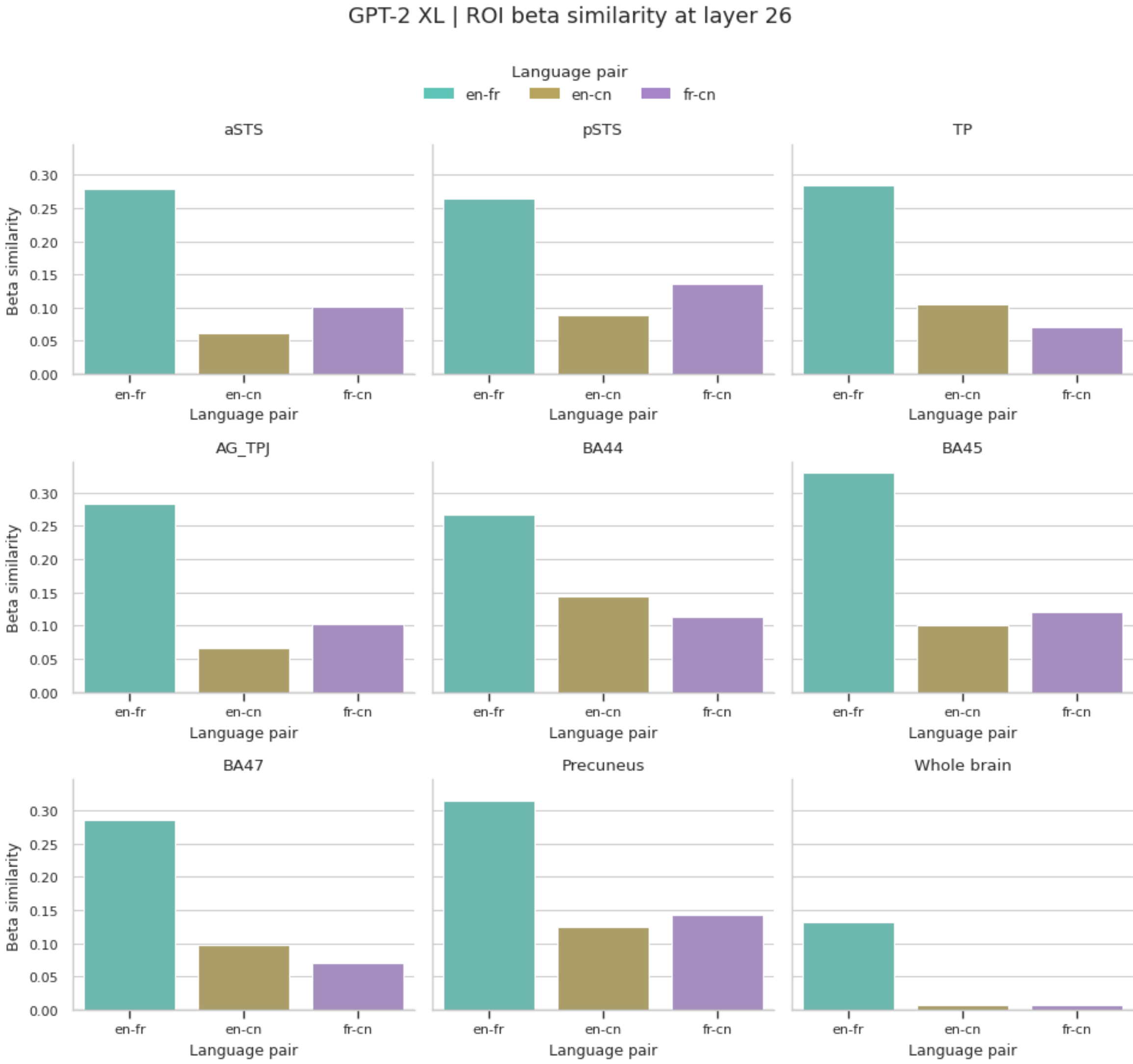


Figure D.4: **ROI-wise beta similarity for GPT-2 XL.** Beta similarity was computed at layer 26 for the three language pairs: English–French, English–Chinese, and French–Chinese. GPT-2 XL shows low beta similarity overall, with English–French similarity higher than pairs involving Chinese.

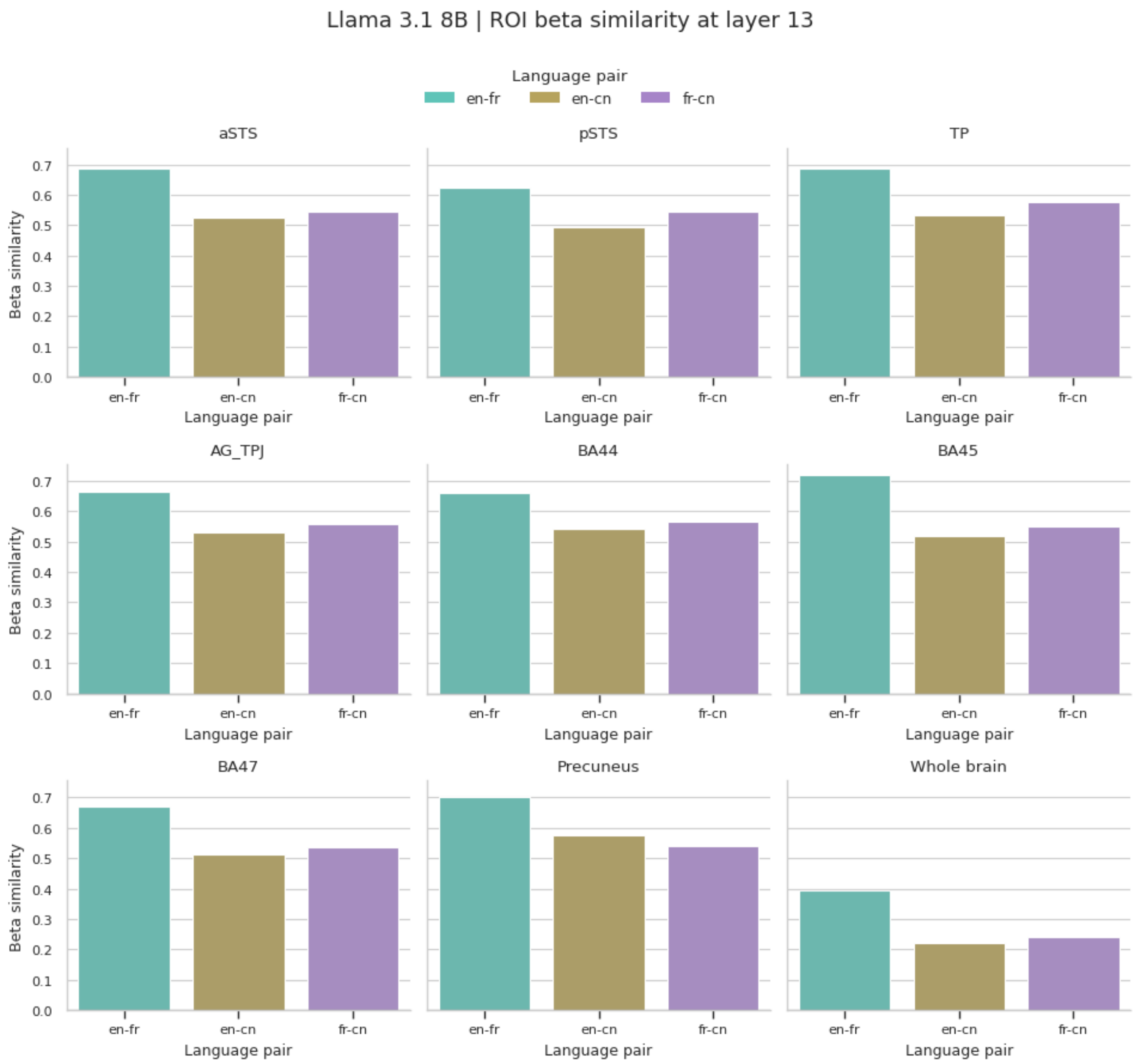


Figure D.5: **ROI-wise beta similarity for Llama 3.1 8B.** Beta similarity was computed at layer 13. Llama 3.1 8B shows substantially higher beta similarity across ROIs than GPT-2 XL, with positive similarity for all three language pairs.

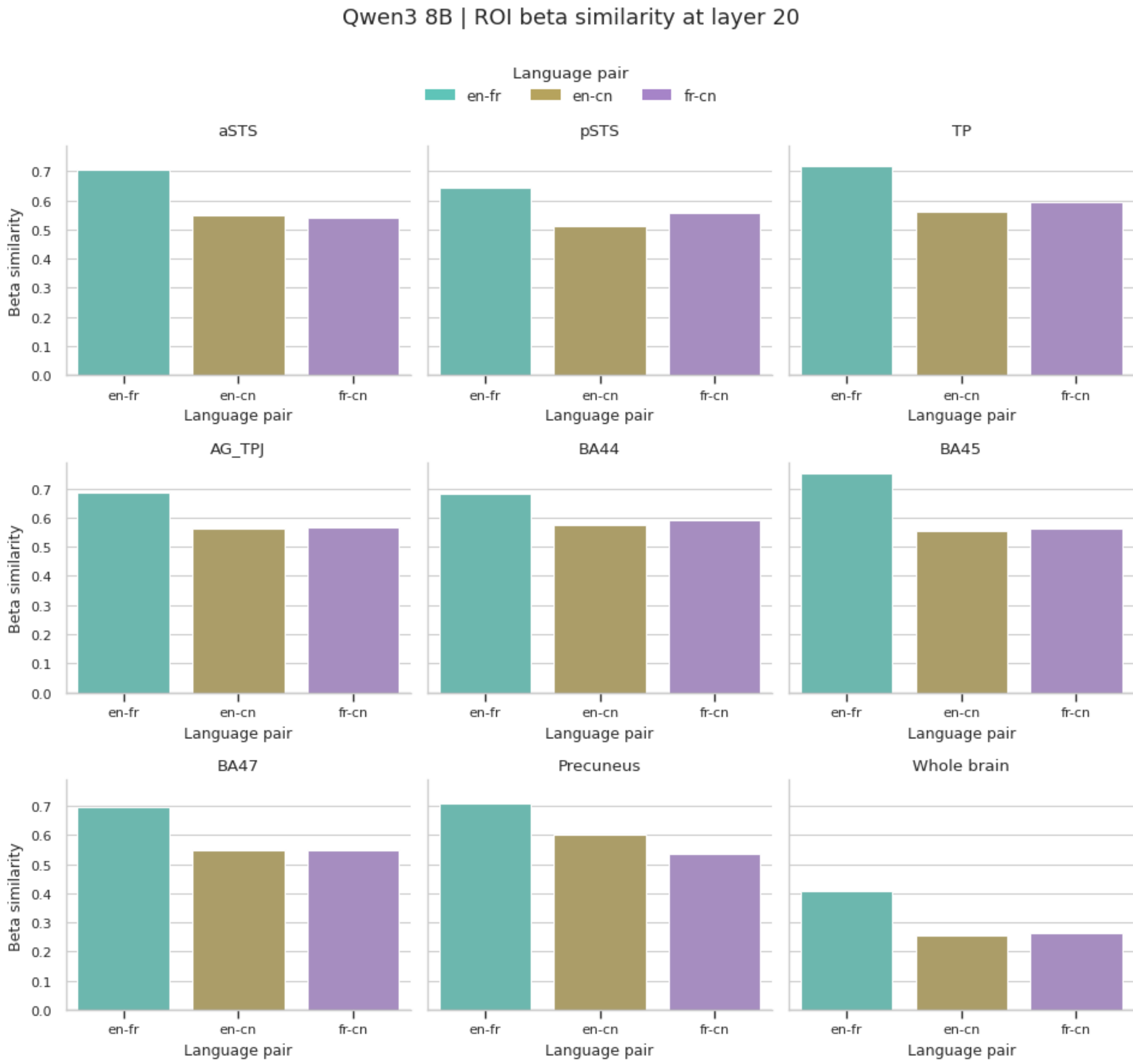


Figure D.6: **ROI-wise beta similarity for Qwen3 8B.** Beta similarity was computed at layer 20. Qwen3 8B shows high ROI-wise beta similarity across languages, with English–French generally highest and the two pairs involving Chinese also clearly positive.